

MULTIPLICATIVITY OF THE SYNTHETIC TATE FILTRATION

WEI YANG

ABSTRACT. We resolve the two monoidality conjectures for the synthetic cellular Tate filtration of Annala and Pstrągowski. The perfect-even module filtration has no unital lax monoidal structure for diagonal tensor products, but the synthetic Tate filtration admits a lax $\mathbb{E}_2$ -monoidal structure compatible with realization. The positive construction uses change of rings for modules with even complex bordism homology and a local monoidal refinement on bounded-below modules with flat even complex bordism homology. We obtain an $\mathbb{E}_2$ comparison of the Antieau, Bhatt–Lurie, and Raksit HKR bifiltrations and identify their double associated graded with a derived exterior-power Laurent algebra. We also identify neutral décalage with a Whitehead tower and prove a completed relative Künneth theorem for arbitrary inputs. The associative refinement reconstructs module categories. An explicit pair of filtered algebras has the same underlying filtration and filtration-zero algebra, but the Grothendieck groups of their compact module categories have different ranks. The coefficient $\mathbb{E}_2$ -algebra admits no $\mathbb{E}_3$ refinement.

CONTENTS

1. Introduction	1
2. Change of rings for even complex bordism homology	4
3. An obstruction to global diagonal monoidality	7
4. Multiplication on the synthetic Tate filtration	10
5. Coordinatewise Tate multifiltrations for torus actions	15
6. Multiplicative connective covers	16
7. Multiplicative comparison of HKR filtrations	18
8. Décalage of the synthetic Tate filtration	23
9. Coefficients and relative tensor products	24
10. Reconstruction of module categories	27
11. Filtered algebras with different module categories	29
12. An obstruction to higher refinements	32
References	33

1. INTRODUCTION

For a spectrum with circle action, the cellular filtration of $BS^1 \simeq \mathbb{CP}^\infty$ gives a filtration of its Tate construction. Annala and Pstrągowski equip this filtration with a lax $\mathbb{E}_2$ -monoidal structure [AP, Theorem 3.4]. They also construct a synthetic version and use it to compare three HKR filtrations on cyclic homology. Conjecture 6.12 of [AP] asks for two multiplicative structures: diagonal lax symmetric monoidality of the perfect-even module filtration, and lax $\mathbb{E}_2$ -monoidality of the synthetic Tate filtration. The first assertion is false, whereas the second holds.

We use their notation for the synthetic sphere and the even circle:

$$\mathbb{S}_{\text{syn}} = \text{fil}_{\text{pev}, \mathbb{S}}(\mathbb{S}), \quad \mathbf{T}_{\text{ev}} = \text{fil}_{\text{pev}, \mathbb{S}[S^1]}(\mathbb{S}[S^1]), \quad \text{Syn} = \text{Mod}_{\mathbb{S}_{\text{syn}}}(\text{Fil}(\text{Sp})).$$

Here $\mathrm{fil}_{\mathrm{pev},B}$ is the perfect-even filtration of B -modules [Pst, Definition 2.21]. The even circle is a bicommutative bialgebra over $\mathbb{S}_{\mathrm{syn}}$ [AR, Corollary 2.44]. Its module category therefore carries the diagonal tensor product: the underlying tensor product is over $\mathbb{S}_{\mathrm{syn}}$, and the action is restricted along the coproduct of $\mathbf{T}_{\mathrm{ev}}$. We denote this tensor product by $\otimes_{\Delta}$.

Let $\mathbb{C}$ denote the one-dimensional complex representation of S^1 of weight one. The filtered representation spheres $\{S^{-q\mathbb{C}}\}_q$ give a filtered $\mathbf{T}_{\mathrm{ev}}$ -module

$$S_{\mathrm{ev}}^q = \mathrm{fil}_{\mathrm{pev},\mathbb{S}[S^1]}(S^{-q\mathbb{C}}).$$

For a $\mathbf{T}_{\mathrm{ev}}$ -module X , write

$$F_T^q(X^{t\mathbf{T}_{\mathrm{ev}}}) = (S_{\mathrm{ev}}^q \otimes_{\Delta} X)^{h\mathbf{T}_{\mathrm{ev}}}. \quad (1)$$

The notation on the left denotes a functor of X , as in [AP, Definition 6.10 and Warning 6.11]. Fixed points here mean the internal mapping object $\mathrm{map}_{\mathbf{T}_{\mathrm{ev}}}(\mathbb{S}_{\mathrm{syn}}, -)$.

Theorem 1.1. (i) *The perfect-even module filtration*

$$\mathrm{fil}_{\mathrm{pev},\mathbb{S}[S^1]} : (\mathrm{Mod}_{\mathbb{S}[S^1]}(\mathrm{Sp}), \otimes_{\Delta}) \longrightarrow (\mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn}), \otimes_{\Delta})$$

admits no unital lax monoidal structure. In particular, Conjecture 6.12(1) of [AP] is false.

(ii) *The synthetic Tate filtration admits a natural lax $\mathbb{E}_2$ -monoidal refinement*

$$F_T^*(-^{t\mathbf{T}_{\mathrm{ev}}}) : (\mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn}), \otimes_{\Delta}) \longrightarrow \mathrm{Fil}(\mathrm{Syn}).$$

For this refinement the canonical realization comparison

$$\mathrm{Re}(F_T^*(X^{t\mathbf{T}_{\mathrm{ev}}})) \longrightarrow F_T^*((\mathrm{Re} X)^{tS^1})$$

is lax $\mathbb{E}_2$ -monoidal. Here Re takes the colimit in the synthetic direction. Thus Conjecture 6.12(2) of [AP] holds.

The obstruction is detected by the regular circle module and its augmentation. After rationalization, naturality and the two unit axioms force an invertible linear map on $\mathbb{Q}^2$ to factor through $\mathbb{Q}$. Section 3 gives the full argument; it excludes even nonsymmetric unital lax structures.

The positive result requires multiplication on the particular coefficient object S_{ev}^* . More generally, a module whose underlying spectrum has even MU -homology has the same perfect-even filtration after restriction to the sphere. Descent along complex bordism proves this comparison after completion; exhaustivity then removes the completion. For bounded-below modules of this kind, the filtration is complete and agrees with the Hahn–Raksit–Wilson filtration. The latter has coherent external tensor maps on algebra–module pairs. Inverting the restriction-of-scalars comparisons for diagonals turns these maps into the required diagonal tensor maps on a tensor-closed subcategory. In particular, this transports the $\mathbb{E}_2$ -algebra of representation spheres constructed in [AP, Theorem 3.6] to S_{ev}^* . Applying fixed points gives the positive part of theorem 1.1.

The local construction extends to arbitrary finite-dimensional tori and bounded-below modules with flat even MU -homology (Corollary 4.7). It includes arbitrary torus actions on finite even-cell spectra and their retracts. Flatness ensures tensor closure; even MU -homology alone does not, as the Moore spectrum example in Remark 4.9 shows.

The same construction is compatible with realization. This supplies the multiplicative input for the following refinement of [AP, Theorem 7.9]. Let k be a classical commutative ring and A an animated commutative k -algebra. The symbols F_{Ant} , F_{BhLu} , and F_{Rak} denote the HKR filtrations of Antieau, Bhatt–Lurie, and Raksit [Ant19, BL, Rak], with the indexing of [AP, Section 7]. Write $\mathrm{BiFil}^c(D(k))$ for bifiltered complexes complete in both directions, with completed Day tensor product, and let $\sigma(r, s) = (s, r)$.

Theorem 1.2. *There are natural equivalences*

$$\begin{aligned} F_{\text{Ant}}^* F_T^* \text{HP}(A/k) &\simeq F_{\text{BhLu}}^* F_T^* \text{HP}(A/k) \\ &\simeq \sigma^* F_T^* F_{\text{Rak}}^* \text{HP}(A/k) \end{aligned}$$

in $\text{Alg}_{\mathbb{E}_2}(\text{BiFil}^c(D(k)))$. At Tate degree zero these induce equivalences of the three filtered $\mathbb{E}_2$ -algebras on $\text{HC}^-(A/k)$.

The underlying equivalences are those of Annala and Pstrągowski. Our assertion concerns their multiplicative refinements with the structures specified in the proof. On polynomial algebras, the comparison maps exhibit Beilinson connective covers. The universal property of connective truncation then determines their operadic compatibility. Sifted left Kan extension in complete bifiltered objects gives the statement for animated algebras.

Let $L_{A/k}$ denote the cotangent complex and Λ_A^m its derived exterior powers. Theorem 7.4 identifies the double associated graded, including its specified $\mathbb{E}_2$ structure, with

$$\left(\bigoplus_{m \geq 0} (\Lambda_A^m L_{A/k})[m] \right) [u^{\pm 1}].$$

The exterior summand has bidegree $(m, 0)$, while u has bidegree $(-1, 1)$ and homological degree -2 . On polynomial algebras the binary multiplication is the wedge–Laurent multiplication, and every component is concentrated in homological degree $r - s$. This purity determines all operadic coherences. Sifted extension then gives a natural $\mathbb{E}_\infty$ refinement of the double associated graded for animated algebras.

We also compare the synthetic Tate filtration with a Whitehead tower. The neutral t -structure on Syn is induced by the levelwise t -structure on filtered spectra; it induces a t -structure on $\mathbf{T}_{\text{ev}}$ -modules [AR, Example 2.77 and Remark 2.78]. Denote its connective covers by $\tau_{\geq a}^N$. For a filtration F , let $d_! F$ be its doubling, $(d_! F)^q = F^{\lceil q/2 \rceil}$, and let Dec_N denote décalage for the neutral Beilinson t -structure: its a th term is the colimit of the Beilinson a -connective cover of F .

Theorem 1.3. *For every $X \in \text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$, there is a natural equivalence*

$$\text{Dec}_N(d_! F_T^*(X^{t\mathbf{T}_{\text{ev}}})) \simeq (\tau_{\geq *}^N X)^{t\mathbf{T}_{\text{ev}}}.$$

Consequently the functor on the left admits a lax symmetric monoidal refinement.

The last refinement is transported from the Whitehead tower followed by the synthetic Tate construction.

Put $\mathcal{R}_T = F_T^*(S_{\text{syn}}^{t\mathbf{T}_{\text{ev}}})$, with trivial action on S_{syn} . The associative part of the specified multiplication makes every value of F_T^* an $\mathcal{R}_T$ -bimodule. Write $\text{Fil}^c(\text{Syn})$ for complete filtrations with completed Day tensor product.

Theorem 1.4. *For all $X, Y \in \text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$, the canonical comparison is an equivalence*

$$F_T^*(X^{t\mathbf{T}_{\text{ev}}}) \hat{\otimes}_{\mathcal{R}_T} F_T^*(Y^{t\mathbf{T}_{\text{ev}}}) \xrightarrow{\sim} F_T^*((X \otimes_\Delta Y)^{t\mathbf{T}_{\text{ev}}})$$

of $\mathcal{R}_T$ -bimodules in $\text{Fil}^c(\text{Syn})$. The tensor product uses the right action on the first factor and the left action on the second. Completion is unnecessary if one input is compact.

The coefficient multiplication is periodic: tensoring with S_{ev}^a translates the external filtration by a . This proves the relative comparison on coefficient spheres. Their Euler cofiber gives $\mathbf{T}_{\text{ev}}^\vee$, hence the free module $\mathbf{T}_{\text{ev}}$ after the appropriate suspension and internal shift. Trivial synthetic coefficients then give compact generators, and completed colimits give the theorem for all inputs. In particular, the filtered Tate images of virtual complex representation spheres are invertible bimodules and satisfy the tensor-power formula of Corollary 9.5.

The associative refinement also reconstructs module categories. Here and below RMod denotes right modules.

Theorem 1.5. *Let B be an $\mathbb{E}_1$ -algebra in $\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$, and put $S = F_T^*(B^{t\mathbf{T}_{\text{ev}}})$. The specified multiplication induces an equivalence*

$$\text{RMod}_B(\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})) \xrightarrow{\sim} \text{RMod}_S(\text{Fil}^c(\text{Syn})), \quad M \mapsto F_T^*(M^{t\mathbf{T}_{\text{ev}}}).$$

For $B = \mathbb{S}_{\text{syn}}$, the proof identifies coefficient spheres with free filtered coefficient modules and uses the Euler cofiber to generate the source. The completed Künneth comparison then identifies the free-module monads for general B .

Theorem 11.2 gives a concrete application. We construct two filtered synthetic algebras $\mathcal{S}_0$ and $\mathcal{S}_1$ with equivalent underlying filtered objects and equivalent algebras in Tate degree zero. Both their stagewise realizations and their external colimits vanish. Nevertheless, writing $\mathcal{M}_i$ for the category of right $\mathcal{S}_i$ -modules in $\text{Fil}^c(\text{Syn})$ and $\mathcal{M}_i^\omega$ for its compact objects, we obtain

$$K_0(\mathcal{M}_0^\omega) \cong \mathbb{Z}^2, \quad K_0(\mathcal{M}_1^\omega) \cong \mathbb{Z}.$$

The calculation transfers two explicit endomorphism algebras through theorem 1.5, detecting multiplication between nonzero filtration degrees.

Proposition 12.3 rules out an $\mathbb{E}_3$ refinement of the coefficient algebra S_{ev}^* , and Proposition 12.5 gives a related obstruction for synthetic Tate filtrations with their canonical filtration-zero fixed-point structure. A filtered Dyer–Lashof operation detects the obstruction.

Conventions. All categories are ∞ -categories. Filtrations are decreasing and integer-indexed:

$$\text{Fil}(\mathcal{C}) = \text{Fun}(\mathbb{Z}^{\text{op}}, \mathcal{C}), \quad \text{gr}^q F = \text{cofib}(F^{q+1} \rightarrow F^q).$$

They carry Day convolution. We write $|F| = \text{colim}_{q \rightarrow -\infty} F^q$, and say that F is complete if $\lim_{q \rightarrow +\infty} F^q = 0$. The notation $\text{Fil}^c(\mathcal{C})$ denotes complete filtrations with completed Day tensor product. The notation $\text{ins}^a(V)$ denotes the free decreasing filtration generated by V in degree a : its value is V in degrees $q \leq a$ and zero in degrees $q > a$. The notation $[m]$ means suspension by m , so that $\pi_i(X[m]) = \pi_{i-m}X$. The internal filtration shift of a synthetic spectrum is $(X(m))^r = X^{r-m}$; it is distinct from $[m]$. An external Tate degree is always indexed by q or s , and the internal degree by r when both occur.

The symbol $\mathbb{S}[S^1] = \Sigma_+^\infty S^1$ denotes the circle group algebra throughout. We use the diagonal tensor product on circle modules and on even-circle modules. Tensor products over a ring, including those used in descent, have that ring written as a subscript. The symbol $L_{A/k}$ denotes the cotangent complex. Exterior powers $\Lambda^q L_{A/k}$ are the usual derived exterior powers, with $\Lambda^q = 0$ for $q < 0$. In particular, $\text{gr}_{\text{Rak}}^q \text{HH}(A/k) \simeq (\Lambda^q L_{A/k})[q]$.

2. CHANGE OF RINGS FOR EVEN COMPLEX BORDISM HOMOLOGY

For a map $f: B \rightarrow C$ of $\mathbb{E}_\infty$ -rings and a C -module N , the morphism of algebra–module pairs (f, id_N) induces a restriction comparison

$$\kappa_{f,N}^{\text{pev}}: \text{fil}_{\text{pev},B}^*(f^*N) \rightarrow \text{Res}_{\text{fil}_{\text{pev}}(f)} \text{fil}_{\text{pev},C}^*(N).$$

Here $\text{fil}_{\text{pev}}^*(f)$ denotes the map between the perfect-even filtrations of the rings themselves. These maps are natural in algebra–module pairs [Pst, Theorem 3.37 and Construction 6.11]. In particular, restriction comparisons for composite ring maps compose with their canonical higher coherences.

Proposition 2.1. *Let B be a connective $\mathbb{E}_\infty$ -ring and let M be a B -module such that $MU_{2j+1}(UM) = 0$ for every $j \in \mathbb{Z}$. The unit of B induces an equivalence*

$$c_{B,M}: \text{fil}_{\text{pev},\mathbb{S}}^*(UM) \xrightarrow{\sim} \text{Res}_{\mathbb{S}_{\text{syn}}} \text{fil}_{\text{pev},B}^*(M)$$

of $\mathbb{S}_{\text{syn}}$ -modules. On the right, restriction uses the ring map $\mathbb{S}_{\text{syn}} \rightarrow \text{fil}_{\text{pev},B}^*(B)$ induced by $\mathbb{S} \rightarrow B$.

The proof compares two varying-base descent diagrams. Their terms agree because the corresponding modules have even homotopy groups. This first identifies the completed filtrations; exhaustivity then identifies the original filtrations.

Proof. The map is $\mathbb{S}_{\text{syn}}$ -linear by functoriality on algebra–module pairs. It suffices to prove that it is an equivalence on underlying filtered spectra. The map $\mathbb{S} \rightarrow MU$ is evenly free in the sense of Hahn–Raksit–Wilson [HRW, Proposition 2.2.20]. Evenly free maps remain evenly free after arbitrary base change [HRW, proof of Corollary 2.2.21]. Thus both $\mathbb{S} \rightarrow MU$ and $B \rightarrow B \otimes MU$ are evenly faithfully flat. Their sources and targets are connective, so they are faithfully even flat in the sense needed for Pstragowski’s descent theorem [Pst, Proposition 6.19].

Applying varying-base descent [Pst, Theorem 6.27] to these two maps gives descent maps

$$\begin{aligned} \text{fil}_{\text{pev}, \mathbb{S}}^*(UM) &\longrightarrow \text{Tot}_{[n] \in \Delta} \text{fil}_{\text{pev}, MU^{\otimes(n+1)}}^*(MU^{\otimes(n+1)} \otimes UM), \\ U \text{fil}_{\text{pev}, B}^*(M) &\longrightarrow \text{Tot}_{[n] \in \Delta} U \text{fil}_{\text{pev}, B \otimes MU^{\otimes(n+1)}}^*(MU^{\otimes(n+1)} \otimes M). \end{aligned}$$

They become equivalences after completion. The map $c_{B,M}$ and the restriction comparisons in each cosimplicial degree give a natural map between these descent diagrams. In the second line we used the canonical identifications

$$(B \otimes MU)^{\otimes_B(n+1)} \simeq B \otimes MU^{\otimes(n+1)}, \quad (B \otimes MU)^{\otimes_B(n+1)} \otimes_B M \simeq MU^{\otimes(n+1)} \otimes M.$$

The calculation in [HRW, proof of Proposition 2.2.20] identifies $\pi_*(MU \otimes MU)$ as a free MU_* -module on even-degree generators. Choosing representatives of a homogeneous basis exhibits $MU \otimes MU$, as a module over either factor, as a direct sum of even suspensions of MU . If an MU -module has even homotopy, smashing it over $\mathbb{S}$ with another copy of MU therefore preserves this parity. Induction gives

$$\pi_{2j+1}(MU^{\otimes(n+1)} \otimes UM) = 0 \quad (n \geq 0, j \in \mathbb{Z}).$$

Thus the module underlying both terms in cosimplicial degree n has even homotopy. By [Pst, Lemma 2.36], both filtered spectra are its double-speed Postnikov filtration.

These identifications identify the comparison map itself with an equivalence. To see this, both filtrations in degree q are $2q$ -connective covers of the same spectrum, and the restriction comparison commutes with their maps to that spectrum. The universal property of a connective cover identifies this map with the identity under the Postnikov identifications. This argument is natural, so the termwise equivalences respect the cosimplicial structure. The map between totalizations is therefore an equivalence. The naturality square for descent now shows that $c_{B,M}$ becomes an equivalence after completion.

Let D^* be the cofibre of $c_{B,M}$. Completion does not change associated graded objects: the fibre of a filtration’s map to its completion is the constant filtration on its limit. Hence $\text{gr } D^* = 0$, so every transition map of D^* is an equivalence and D^* is constant. Perfect-even filtrations are exhaustive, and the colimit of a restriction comparison is the identity of the underlying module [Pst, Proposition 2.23 and Remark 6.12]. It follows that $\text{colim } D^* = 0$. A constant filtration with zero colimit is zero, which proves the proposition. $\square$

Corollary 2.2. *Let $f: B \rightarrow C$ be a map of connective $\mathbb{E}_\infty$ -rings, and let N be a C -module whose underlying spectrum has even MU -homology. Then $\kappa_{f,N}^{\text{pev}}$ is an equivalence. If UN is bounded below, $\text{fil}_{\text{pev}, B}^*(f^*N)$ and $\text{fil}_{\text{pev}, C}^*(N)$ are complete. In particular, both assertions apply when UN is an even-dimensional sphere.*

Proof. The comparisons from $\mathbb{S}$ to B and C are equivalences by Proposition 2.1. Their composition law identifies the latter with the former followed by $\kappa_{f,N}^{\text{pev}}$. The two-out-of-three property proves the first assertion.

Even MU -homology is equivalent to homological evenness over $\mathbb{S}$ [Pst, Theorem 4.16 and Warning 6.10]. For bounded-below UN , the completeness theorem [Pst, Theorem 8.3]

applies after an even suspension making it connective. The identity

$$\mathrm{fil}_{\mathrm{pev}, \mathbb{S}}^q(\Sigma^{2d} X) \simeq \Sigma^{2d} \mathrm{fil}_{\mathrm{pev}, \mathbb{S}}^{q-d}(X)$$

follows from the definition of the perfect-even filtration, so an even suspension preserves completeness. Proposition 2.1 then gives completeness over both B and C . $\square$

2.1. Comparison with the module even filtration. We use the Hahn–Raksit–Wilson filtration to identify the resulting module structures with the prescribed bialgebra structure on $\mathbf{T}_{\mathrm{ev}}$. Write $\mathrm{fil}_{\mathbb{E}_{\infty}\text{-ev}, B}^*(M)$ for this module even filtration. Its description as a right Kan extension gives

$$\mathrm{fil}_{\mathbb{E}_{\infty}\text{-ev}, B}^*(M) \simeq \lim_{B \rightarrow C, \pi_* C \text{ even}} \tau_{\geq 2*}(C \otimes_B M) \quad (2)$$

[HRW, Construction A.1.3 and Remark A.1.4].

Lemma 2.3. *The comparison of [Pst, Construction 7.4] refines naturally to a map of algebra–module pairs*

$$(\mathrm{fil}_{\mathrm{pev}, B}^*(B), \mathrm{fil}_{\mathrm{pev}, B}^*(M)) \longrightarrow (\mathrm{fil}_{\mathbb{E}_{\infty}\text{-ev}, B}^*(B), \mathrm{fil}_{\mathbb{E}_{\infty}\text{-ev}, B}^*(M)).$$

In particular, its module component is linear over its ring component.

Proof. For every $B \rightarrow C$ with $\pi_* C$ even, functoriality on algebra–module pairs gives a map to

$$(\mathrm{fil}_{\mathrm{pev}, C}^*(C), \mathrm{fil}_{\mathrm{pev}, C}^*(C \otimes_B M)).$$

By [Pst, Proposition 2.40], this pair is canonically its double-speed Postnikov filtration. This identification respects the ring and module operations: after evaluation in filtration degrees, the operations are the unique lifts of the original operations through the appropriate connective covers. The maps therefore form a cone of algebra–module pairs over the diagram in (2). Limits of such pairs are computed on the underlying ring and module objects, with the induced action. Taking the limit gives the asserted refinement. After forgetting structure, this is precisely Construction 7.4, since both comparisons restrict to the canonical Postnikov identification over even rings. All maps in the cone are natural in algebra–module pairs, which also proves the stated naturality. $\square$

Proposition 2.4. *For $r \geq 0$, the canonical ring comparison*

$$\mathrm{fil}_{\mathrm{pev}, \mathbb{S}[(S^1)^r]}^*(\mathbb{S}[(S^1)^r]) \xrightarrow{\sim} \mathrm{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[(S^1)^r]}^*(\mathbb{S}[(S^1)^r])$$

is an equivalence. For every $\mathbb{S}[(S^1)^r]$ -module M whose underlying spectrum is bounded below and has even MU -homology, the canonical map

$$\mathrm{fil}_{\mathrm{pev}, \mathbb{S}[(S^1)^r]}^*(M) \xrightarrow{\sim} \mathrm{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[(S^1)^r]}^*(M)$$

is an equivalence of modules over this ring comparison. Here $r = 0$ means $\mathbb{S}[(S^1)^0] = \mathbb{S}$.

Proof. The augmentation $B[S^1] \rightarrow B$ is evenly faithfully flat for every $\mathbb{E}_{\infty}$ -ring B [AR, Corollary 2.36]. This property is closed under composition: after base change to an even ring, each successive map is an even faithfully flat extension, and their composite is faithfully flat. Iterating the augmentation and then using $\mathbb{S} \rightarrow MU$ shows that

$$\mathbb{S}[(S^1)^r] \longrightarrow \mathbb{S} \longrightarrow MU$$

is evenly faithfully flat. Both endpoints are connective, so [Pst, Proposition 6.19] identifies it as faithfully even flat. Since MU is even, [Pst, Theorem 7.5] identifies each comparison in the statement as completion.

For the modules in the statement, completeness is Corollary 2.2. The ring $\mathbb{S}[(S^1)^r]$ is connective and is perfect even as a module over itself; its perfect-even filtration is therefore complete by [Pst, Lemma 2.18 and Theorem 8.3]. Finally, Lemma 2.3 supplies module linearity. $\square$

Corollary 2.5. *For any map $f: \mathbb{S}[(S^1)^s] \rightarrow \mathbb{S}[(S^1)^t]$ of $\mathbb{E}_\infty$ -rings and any $\mathbb{S}[(S^1)^t]$ -module N whose underlying spectrum is bounded below and has even MU-homology, the canonical map*

$$\kappa_{f,N}^{\text{ev}}: \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}[(S^1)^s]}^*(f^*N) \xrightarrow{\sim} \text{Res}_{\text{fil}_{\mathbb{E}_\infty\text{-ev}}^*(f)} \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}[(S^1)^t]}^*(N)$$

is an equivalence. These equivalences are natural in algebra–module pairs and compatible with composition, including all higher coherences.

Proof. Naturality in Lemma 2.3 gives a square whose horizontal arrows are the perfect-even/HRW comparisons and whose vertical arrows are the two restriction comparisons. The horizontal arrows are equivalences by Proposition 2.4, and the perfect-even restriction comparison is an equivalence by Corollary 2.2. Thus the HRW restriction comparison is an equivalence. Its coherence is inherited from the functor on algebra–module pairs defining $\text{fil}_{\mathbb{E}_\infty\text{-ev}}^*$. $\square$

3. AN OBSTRUCTION TO GLOBAL DIAGONAL MONOIDALITY

Theorem 3.1. *The perfect-even module filtration*

$$\text{fil}_{\text{pev}, \mathbb{S}[S^1]}: (\text{Mod}_{\mathbb{S}[S^1]}(\text{Sp}), \otimes_\Delta) \longrightarrow (\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn}), \otimes_\Delta)$$

does not admit a unital lax monoidal structure. In particular, Conjecture 6.12(1) of [AP] is false.

The obstruction uses the regular circle module, its augmentation, and the unit axioms. After rationalization, those axioms would force a rank-two map to factor through a one-dimensional vector space. We first establish the suspension formula needed to calculate that intermediate space.

Lemma 3.2. *For every $\mathbb{E}_1$ -ring B , there is a natural equivalence of filtered spectra*

$$\text{fil}_{\text{pev}, B}(\Sigma B) \simeq \Sigma \text{fil}_{\text{pev}, B}(B).$$

The perfect-even module filtration also preserves finite direct sums and commutes with termwise rationalization.

Proof. Write $Y_B(M) = \text{map}_B(-, M)$ for the spectral Yoneda sheaf on perfect even B -modules. Pstrągowski’s definition is

$$\text{fil}_{\text{pev}, B}^q(M) = \Gamma(B, \tau_{\geq 2q} Y_B(M)),$$

where the connective cover is taken in sheaves of spectra [Pst, Definition 2.21]. The module B is homologically even [Pst, Lemma 2.18]; equivalently, the odd homotopy sheaves of $Y_B(B)$ vanish. Thus

$$\tau_{\geq 2q-1} Y_B(B) \simeq \tau_{\geq 2q} Y_B(B).$$

The spectral Yoneda functor and the sections functor are exact, so

$$\begin{aligned} \text{fil}_{\text{pev}, B}^q(\Sigma B) &\simeq \Gamma(B, \tau_{\geq 2q} \Sigma Y_B(B)) \\ &\simeq \Sigma \Gamma(B, \tau_{\geq 2q-1} Y_B(B)) \simeq \Sigma \text{fil}_{\text{pev}, B}^q(B). \end{aligned}$$

These equivalences commute with the filtration transitions and keep the filtration index fixed; compare [Pst, Remark 2.35].

Each of Y_B , $\tau_{\geq 2q}$, and $\Gamma(B, -)$ preserves finite products. Since the ambient categories are stable, their composite preserves finite direct sums and is additive. It therefore carries multiplication by an integer to multiplication by the same integer. Rationalization is a filtered colimit of such multiplication maps. The perfect-even filtration preserves filtered colimits [Pst, Lemma 2.24], giving the last assertion. $\square$

For $r \in \mathbb{Z}$, let $\mathbb{Q}(r)$ denote the filtered rational spectrum with value $H\mathbb{Q}$ in degrees $q \leq r$, value zero in degrees $q > r$, and identity transition maps wherever both terms are nonzero. Thus $\mathbb{Q}(0)$ is the Day-convolution unit and

$$\mathbb{Q}(r) \otimes \mathbb{Q}(s) \simeq \mathbb{Q}(r+s).$$

All tensor products between filtered rational spectra in this section are Day convolution over $\mathbb{Q}$.

Lemma 3.2 and [Pst, Lemma 2.36] give the canonical equivalence of filtered rings

$$(\mathbb{S}_{\text{syn}})_{\mathbb{Q}} \simeq \text{fil}_{\text{pev}, \mathbb{S}}(H\mathbb{Q}) \simeq \tau_{\geq 2*} H\mathbb{Q} = \mathbb{Q}(0). \quad (3)$$

The subscript $\mathbb{Q}$ here means termwise rationalization. The comparison respects ring structures by the functoriality of the perfect-even filtration and the canonical multiplicative structure of connective covers. Define the functor

$$\Phi_{\mathbb{Q}}(M) := \mathbb{Q}(0) \otimes_{\mathbb{S}_{\text{syn}}} U \text{fil}_{\text{pev}, \mathbb{S}[S^1]}(M) \in \text{Fil}(\text{Mod}_{H\mathbb{Q}}(\text{Sp})), \quad (4)$$

where U forgets the $\mathbf{T}_{\text{ev}}$ -action. By (3), this is equivalently the underlying perfect-even filtration rationalized termwise. Forgetting a diagonal action and extension of scalars are strong symmetric monoidal. Hence a unital lax monoidal structure in Theorem 3.1 would induce one on $\Phi_{\mathbb{Q}}$.

Lemma 3.3. *There are equivalences*

$$\begin{aligned} \Phi_{\mathbb{Q}}(\mathbb{S}) &\simeq \mathbb{Q}(0), \\ \Phi_{\mathbb{Q}}(\mathbb{S}[S^1]) &\simeq \mathbb{Q}(0) \oplus \Sigma \mathbb{Q}(1), \\ \Phi_{\mathbb{Q}}(\mathbb{S}[S^1] \otimes_{\Delta} \mathbb{S}[S^1]) &\simeq \mathbb{Q}(0) \oplus \Sigma \mathbb{Q}(1) \oplus \Sigma \mathbb{Q}(0) \oplus \Sigma^2 \mathbb{Q}(1). \end{aligned}$$

Under the first two identifications, the image of the augmentation $\epsilon: \mathbb{S}[S^1] \rightarrow \mathbb{S}$ is the projection to $\mathbb{Q}(0)$.

Proof. The augmentation restriction comparison identifies $\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(\mathbb{S})$ with $\mathbb{S}_{\text{syn}}$ by Corollary 2.2. Together with (3), this gives the first equivalence. Antieau–Riggenbach’s split fibre sequence [AR, Lemma 2.40], applied to the sphere, identifies the ring augmentation with the projection in

$$\mathbf{T}_{\text{ev}} \simeq \mathbb{S}_{\text{syn}} \oplus \Sigma \mathbb{S}_{\text{syn}}(1).$$

This is a splitting of synthetic spectra. Naturality on algebra–module pairs identifies the composite

$$\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(\mathbb{S}[S^1]) \longrightarrow \text{fil}_{\text{pev}, \mathbb{S}[S^1]}(\mathbb{S}) \longrightarrow \text{fil}_{\text{pev}, \mathbb{S}}(\mathbb{S})$$

with that ring augmentation. The canonical perfect-even/HRW comparisons respect it by Lemma 2.3. This proves the second equivalence and the assertion about $\Phi_{\mathbb{Q}}(\epsilon)$.

For the third equivalence, the homeomorphism

$$(z, w) \longmapsto (z, z^{-1}w)$$

identifies the diagonal circle action on $S^1 \times S^1$ with the action on its first factor alone. After taking suspension spectra and splitting the second, trivially acted-on circle, this gives an equivalence of $\mathbb{S}[S^1]$ -modules

$$\mathbb{S}[S^1] \otimes_{\Delta} \mathbb{S}[S^1] \simeq \mathbb{S}[S^1] \oplus \Sigma \mathbb{S}[S^1].$$

The result now follows from Lemma 3.2 and the second equivalence. $\square$

Proof of Theorem 3.1. Suppose that the proposed lax monoidal structure exists, and denote the induced structure maps of $\Phi_{\mathbb{Q}}$ by

$$\mu_{M,N}: \Phi_{\mathbb{Q}}(M) \otimes \Phi_{\mathbb{Q}}(N) \longrightarrow \Phi_{\mathbb{Q}}(M \otimes_{\Delta} N).$$

Its unit is a map $\mathbb{Q}(0) \rightarrow \Phi_{\mathbb{Q}}(\mathbb{S}) \simeq \mathbb{Q}(0)$, so its homotopy class is multiplication by a scalar $a \in \mathbb{Q}$. The unit axiom applied to $\Phi_{\mathbb{Q}}(\mathbb{S}[S^1]) \neq 0$ forces $a \neq 0$. Thus the unit map is an equivalence, and the left and right unit axioms give, under the canonical unit identifications,

$$\mu_{\mathbb{S}[S^1], \mathbb{S}} = a^{-1} \text{id}_{\Phi_{\mathbb{Q}}(\mathbb{S}[S^1])}, \quad \mu_{\mathbb{S}, \mathbb{S}[S^1]} = a^{-1} \text{id}_{\Phi_{\mathbb{Q}}(\mathbb{S}[S^1])} \quad (5)$$

in the homotopy category.

Let $p_1 = \text{id} \otimes \epsilon$ and $p_2 = \epsilon \otimes \text{id}$ be the two maps from $\mathbb{S}[S^1] \otimes_{\Delta} \mathbb{S}[S^1]$ to $\mathbb{S}[S^1]$, and put $p = \Phi_{\mathbb{Q}}(\epsilon)$. Naturality and (5) imply

$$\Phi_{\mathbb{Q}}(p_1) \mu_{\mathbb{S}[S^1], \mathbb{S}[S^1]} = a^{-1} (\text{id} \otimes p), \quad (6)$$

$$\Phi_{\mathbb{Q}}(p_2) \mu_{\mathbb{S}[S^1], \mathbb{S}[S^1]} = a^{-1} (p \otimes \text{id}). \quad (7)$$

These two identities already contradict the dimensions in filtration degree one. Indeed, Lemma 3.3 gives

$$\begin{aligned} \pi_1((\Phi_{\mathbb{Q}}(\mathbb{S}[S^1]) \otimes \Phi_{\mathbb{Q}}(\mathbb{S}[S^1]))^1) &\simeq \mathbb{Q}^2, \\ \pi_1(\Phi_{\mathbb{Q}}(\mathbb{S}[S^1] \otimes_{\Delta} \mathbb{S}[S^1])^1) &\simeq \mathbb{Q}. \end{aligned}$$

In the first line, the two basis classes come from $\Sigma\mathbb{Q}(1) \otimes \mathbb{Q}(0)$ and $\mathbb{Q}(0) \otimes \Sigma\mathbb{Q}(1)$; the remaining summand $\Sigma^2\mathbb{Q}(2)$ has zero π_1 . In the second line, only $\Sigma\mathbb{Q}(1)$ contributes to π_1 in filtration degree one.

The two projections on the right sides of (6)–(7) read these basis classes separately, because p is the projection to $\mathbb{Q}(0)$. Combining the identities therefore gives a factorization

$$a^{-1} \text{id}_{\mathbb{Q}^2}: \mathbb{Q}^2 \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}^2.$$

The composite on the right has rank at most one, whereas the map on the left has rank two. This contradiction proves the theorem. $\square$

Corollary 3.4. *For every $d \geq 1$, the functor*

$$\text{fil}_{\text{pev}, \mathbb{S}[(S^1)^d]}: (\text{Mod}_{\mathbb{S}[(S^1)^d]}(\text{Sp}), \otimes_{\Delta}) \longrightarrow (\text{Mod}_{\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}} d}}(\text{Syn}), \otimes_{\Delta}})$$

does not admit a unital lax monoidal structure. The target uses the canonical ring-power equivalence (8) and the bicommutative bialgebra structure induced by the torus diagonal.

Proof. Let $\Phi_{\mathbb{Q}, d}$ be the rationalized functor defined as in (4), with $\mathbb{S}[S^1]$ replaced by $\mathbb{S}[(S^1)^d]$. The ring-power equivalence and the split augmentation of Lemma 3.3 give

$$\Phi_{\mathbb{Q}, d}(\mathbb{S}) \simeq \mathbb{Q}(0), \quad \Phi_{\mathbb{Q}, d}(\mathbb{S}[(S^1)^d]) \simeq (\mathbb{Q}(0) \oplus \Sigma\mathbb{Q}(1))^{\otimes d}.$$

The augmentation is the projection onto $\mathbb{Q}(0)$. Thus the regular module's image has π_1 of dimension d in filtration degree one, and its tensor square has dimension $2d$ there.

The same change of coordinates $(z, w) \mapsto (z, z^{-1}w)$ on the product of two tori, followed by the stable splitting of the second torus, gives

$$\mathbb{S}[(S^1)^d] \otimes_{\Delta} \mathbb{S}[(S^1)^d] \simeq \bigoplus_{j=0}^d (\Sigma^j \mathbb{S}[(S^1)^d])^{\oplus \binom{d}{j}}.$$

After applying $\Phi_{\mathbb{Q}, d}$, only $j = 0$ contributes to π_1 in filtration degree one. For $j = 1$, Lemma 3.2 identifies that group with $\pi_0 \Phi_{\mathbb{Q}, d}(\mathbb{S}[(S^1)^d])^1$, which vanishes by the displayed formula. For $j \geq 2$, the precise bound in [Pst, Proposition 2.23] gives an injection

$$\pi_1 \text{fil}_{\text{pev}, \mathbb{S}[(S^1)^d]}^1(\Sigma^j \mathbb{S}[(S^1)^d]) \hookrightarrow \pi_1(\Sigma^j \mathbb{S}[(S^1)^d]) = 0.$$

Indeed, that proposition gives injectivity in degree $2q - 1$ for filtration degree q , so it applies with $q = 1$. The target vanishes because the torus suspension spectrum is connective and $j \geq 2$. Termwise rationalization preserves this vanishing. Hence

$$\dim_{\mathbb{Q}} \pi_1 \Phi_{\mathbb{Q}, d}(\mathbb{S}[(S^1)^d] \otimes_{\Delta} \mathbb{S}[(S^1)^d])^1 = d.$$

The unit axioms and the two augmentation projections now give exactly the identities (6)–(7) for $\Phi_{\mathbb{Q},d}$. Their right sides jointly read the two independent d -dimensional summands in the tensor square. A putative unit scalar $a \in \mathbb{Q}^\times$ would therefore give a factorization

$$a^{-1} \text{id}_{\mathbb{Q}^{2d}} : \mathbb{Q}^{2d} \longrightarrow \mathbb{Q}^d \longrightarrow \mathbb{Q}^{2d},$$

which is impossible for $d \geq 1$. $\square$

Remark 3.5. The contradiction follows from naturality and the two unit axioms on the regular module and its augmentation. The diagonal tensor product has unit $\mathbb{S}$ with trivial action; the relative tensor product over $\mathbb{S}[S^1]$ has unit $\mathbb{S}[S^1]$. The regular module has underlying spectrum $\mathbb{S} \oplus \Sigma \mathbb{S}$, whereas Proposition 4.1 concerns modules whose underlying spectra are even-dimensional spheres.

4. MULTIPLICATION ON THE SYNTHETIC TATE FILTRATION

The canonical comparisons of Proposition 2.4 identify $\mathbb{S}_{\text{syn}}$ and $\mathbf{T}_{\text{ev}}$ with their Hahn–Raksit–Wilson versions. For the full symmetric monoidal subcategory of rings spanned by $\mathbb{S}[(S^1)^r]$, Antieau–Riggenbach prove that the ring even filtration is strong symmetric monoidal [AR, Proposition 2.43 and proof of Corollary 2.44]. In particular, there are coherent equivalences

$$\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}}} r} \simeq \text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[(S^1)^r]}^*(\mathbb{S}[(S^1)^r]). \quad (8)$$

If $\Delta_r : \mathbb{S}[S^1] \rightarrow \mathbb{S}[(S^1)^r]$ is induced by the diagonal of the circle, its image is the iterated coproduct of $\mathbf{T}_{\text{ev}}$ under (8). For $r = 0$ this means the augmentation. This is the construction of the bialgebra structure in [AR, Corollary 2.44].

Let $\mathcal{C}_{\text{sph}}$ be the full subcategory of $\text{Mod}_{\mathbb{S}[S^1]}(\text{Sp})$ consisting of modules whose underlying spectra are even-dimensional spheres. It contains the trivial-action unit $\mathbb{S}$ and is closed under the diagonal tensor product $\otimes_{\Delta}$.

Proposition 4.1. *The module even filtration restricts to a strong symmetric monoidal functor*

$$\text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[S^1]}^* : (\mathcal{C}_{\text{sph}}, \otimes_{\Delta}) \longrightarrow (\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn}), \otimes_{\Delta}).$$

Proof. The module filtration on algebra–module pairs is lax symmetric monoidal [HRW, Remark A.1.6]. For $M_1, \dots, M_r \in \mathcal{C}_{\text{sph}}$, let $M_1 \boxtimes \dots \boxtimes M_r$ denote their external tensor product over $\mathbb{S}$, a module over $\mathbb{S}[(S^1)^r]$. The external operations of this lax monoidal functor give

$$\begin{aligned} \lambda_r : \text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[S^1]}^*(M_1) \otimes_{\mathbb{S}_{\text{syn}}} \dots \otimes_{\mathbb{S}_{\text{syn}}} \text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[S^1]}^*(M_r) \\ \longrightarrow \text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[(S^1)^r]}^*(M_1 \boxtimes \dots \boxtimes M_r). \end{aligned} \quad (9)$$

The tensor product over $\mathbb{S}_{\text{syn}}$ is obtained by balancing the external operation over the image of the unit ring. Its target is a module over the right side of (8), and λ_r is linear over the corresponding ring tensor map.

Restrict (9) along the coproduct of $\mathbf{T}_{\text{ev}}$. The external product still has an even-dimensional sphere as underlying spectrum. Corollary 2.5, applied to Δ_r , therefore gives an equivalence

$$\kappa_{\Delta_r}^{\text{ev}} : \text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[S^1]}^*(M_1 \otimes_{\Delta} \dots \otimes_{\Delta} M_r) \xrightarrow{\sim} \text{Res}_{\Delta_{\mathbf{T}_{\text{ev}}, r}} \text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[(S^1)^r]}^*(M_1 \boxtimes \dots \boxtimes M_r),$$

where $\Delta_{\mathbf{T}_{\text{ev}}, r}$ is the iterated coproduct of $\mathbf{T}_{\text{ev}}$. Define the desired operation to be

$$\mu_r = (\kappa_{\Delta_r}^{\text{ev}})^{-1} \circ \text{Res}(\lambda_r). \quad (10)$$

For $r = 0$, the same construction uses the augmentation $\mathbb{S}[S^1] \rightarrow \mathbb{S}$ and gives the unit map from the trivial $\mathbf{T}_{\text{ev}}$ -module $\mathbb{S}_{\text{syn}}$.

We check how these operations compose. Restriction of scalars is the cartesian transport in the category of algebra–module pairs over rings: an arrow $(B, f^*N) \rightarrow (C, N)$ over f is cartesian. On the subcategory of pairs with ring $\mathbb{S}[(S^1)^r]$ and underlying even-dimensional sphere, Corollary 2.5 says exactly that the filtration functor carries these cartesian arrows

to cartesian arrows. External tensor products preserve these arrows, since restricting an external product along $f_1 \otimes \cdots \otimes f_r$ is the external product of the restrictions.

Consequently, for any partition of a finite set of inputs, the diagram that compares an external tensor operation before and after restricting each block commutes: it is the naturality diagram of the lax monoidal functor on algebra–module pairs. For successive restrictions the comparison is

$$\kappa_{g \circ f, N}^{\text{ev}} \simeq \text{Res}_{\text{fil}_{\mathbb{E}_\infty\text{-ev}}^*(f)}(\kappa_{g, N}^{\text{ev}}) \circ \kappa_{f, g^* N}^{\text{ev}}.$$

These identities are coherent in algebra–module pairs. All the comparisons are equivalences, and their inverses inherit the same coherences.

Now postcompose either iterated version of (10) associated with a partition by the comparison for the full diagonal. The preceding naturality diagram and composition identity identify the result with the corresponding composite of external operations λ_r . These external operations satisfy the commutative-operad composition and permutation laws. This identification works on the entire diagrams of successive partitions, so it transports their higher homotopies as well. It also includes empty blocks and the nullary operation. Thus (10) supplies the full lax symmetric monoidal structure.

It remains to show that the structure maps are equivalences. For an even-dimensional sphere, the canonical even-suspension identification is

$$\text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}}(S^{2d}) \simeq \mathbb{S}_{\text{syn}}[2d](d).$$

It follows directly from the even-algebra limit (2). These identifications respect the tensor comparisons: at an even test algebra, both the filtration and its products are the corresponding shifts of the Postnikov filtration and its multiplication. Thus the relative tensor comparison over $\mathbb{S}_{\text{syn}}$ on two sphere inputs is the shifted unit multiplication, and is an equivalence.

For circle modules M, N in $\mathcal{C}_{\text{sph}}$, the restriction comparisons along the units of the group rings give a commutative square

$$\begin{array}{ccc} \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}}(UM) \otimes_{\mathbb{S}_{\text{syn}}} \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}}(UN) & \longrightarrow & \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}}(UM \otimes UN) \\ \sim \downarrow & & \downarrow \sim \\ U \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}[S^1]}(M) \otimes_{\mathbb{S}_{\text{syn}}} U \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}[S^1]}(N) & \longrightarrow & U \text{fil}_{\mathbb{E}_\infty\text{-ev}, \mathbb{S}[S^1]}(M \otimes_\Delta N). \end{array}$$

Here the lower arrow is exactly (10). Commutativity follows from naturality of the external operations on algebra–module pairs and composition of restriction comparisons; the unit map followed by the circle diagonal is the unit map of the product group ring. The vertical arrows are equivalences by Corollary 2.5. The upper arrow is the sphere comparison just calculated, so the lower arrow is an equivalence. The same argument applies to the nullary operation. Since the coherences are those already constructed in Proposition 4.1, this proves strong symmetric monoidality. $\square$

4.1. The filtered representation spheres. We recall the part of the Day-convolution construction needed here.

Lemma 4.2. *Let $\mathcal{D} \subseteq \mathcal{C}$ be a tensor-closed full subcategory of a presentably symmetric monoidal category, and let $F: \mathcal{D} \rightarrow \mathcal{E}$ be lax symmetric monoidal, where $\mathcal{E}$ is presentably symmetric monoidal. If an $\mathbb{E}_2$ -algebra K^* in $\text{Fil}(\mathcal{C})$ takes all its values in $\mathcal{D}$, the termwise diagram $F(K^*)$ has a natural $\mathbb{E}_2$ -algebra structure in $\text{Fil}(\mathcal{E})$.*

Proof. For each operation in the little-disks operad and each tuple of filtration indices, compose

$$F(K^{q_1}) \otimes \cdots \otimes F(K^{q_r}) \longrightarrow F(K^{q_1} \otimes \cdots \otimes K^{q_r}) \longrightarrow F(K^{q_1 + \cdots + q_r}).$$

The second arrow is the image of the corresponding operation on K^* . It is defined in $\mathcal{D}$ because that subcategory is full and tensor-closed. These maps are compatible with the filtration transitions, so the defining colimits for Day convolution give operations on $F(K^*)$. Composition of these operations is inherited from the operad action on K^* and the lax symmetric monoidal structure of F ; the same construction applies to their parameter spaces and higher coherences. This is the multimorphism form of the Day-convolution universal property [Gla]. $\square$

Theorem 4.3. *The filtered $\mathbf{T}_{\text{ev}}$ -module*

$$S_{\text{ev}}^* = \{\text{fil}_{\text{pev}, \mathbb{S}[S^1]}^*(S^{-q\mathbb{C}})\}_{q \in \mathbb{Z}} \in \text{Fil}(\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn}))$$

of [AP, Notation 6.9], where the perfect-even filtration is applied termwise, admits an $\mathbb{E}_2$ -algebra structure for diagonal tensor products.

Proof. By [AP, Theorem 3.6], the diagram $q \mapsto S^{-q\mathbb{C}}$ is an $\mathbb{E}_2$ -algebra in $\text{Fil}(\text{Mod}_{\mathbb{S}[S^1]}(\text{Sp}))$ for diagonal tensor products. Its term in degree q has underlying spectrum $\Sigma^{-2q}\mathbb{S}$, so every term belongs to $\mathcal{C}_{\text{sph}}$. Proposition 4.1 and Lemma 4.2 give the termwise HRW filtration of this diagram an $\mathbb{E}_2$ -algebra structure. Proposition 2.4 and Lemma 2.3 give a natural equivalence

$$S_{\text{ev}}^* \xrightarrow{\sim} \{\text{fil}_{\mathbb{E}_{\infty}\text{-ev}, \mathbb{S}[S^1]}^*(S^{-q\mathbb{C}})\}_{q \in \mathbb{Z}}$$

of filtered $\mathbf{T}_{\text{ev}}$ -modules. Transporting the operad action through this equivalence proves the assertion. $\square$

Theorem 4.4. *The synthetic Tate filtration*

$$X \longmapsto F_T^*(X^{t\mathbf{T}_{\text{ev}}}) = (S_{\text{ev}}^* \otimes_{\mathbb{S}_{\text{syn}}} X)^{h\mathbf{T}_{\text{ev}}}$$

admits a lax $\mathbb{E}_2$ -monoidal structure as a functor

$$(\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn}), \otimes_{\Delta}) \longrightarrow \text{Fil}(\text{Syn}).$$

In particular, Conjecture 6.12(2) of [AP] holds.

Proof. Tensoring with an $\mathbb{E}_2$ -algebra $K^* \in \text{Fil}(\mathcal{C})$ defines a lax $\mathbb{E}_2$ -monoidal functor $X \mapsto \{K^q \otimes X\}_q$ from $\mathcal{C}$ to $\text{Fil}(\mathcal{C})$. Its operations are induced by those of K^* , after moving the factors X_i past the factors K^{q_j} using the symmetry. Day convolution assembles these maps as in Lemma 4.2. Apply this to $\mathcal{C} = \text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$ with diagonal tensor product and $K^* = S_{\text{ev}}^*$ from Theorem 4.3.

The trivial-action functor $\text{Syn} \rightarrow \text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$ is strong symmetric monoidal, by the counit identities for the bialgebra $\mathbf{T}_{\text{ev}}$. Its right adjoint $(-)^{h\mathbf{T}_{\text{ev}}} = \text{map}_{\mathbf{T}_{\text{ev}}}(\mathbb{S}_{\text{syn}}, -)$ is therefore lax symmetric monoidal. Applying it termwise carries the preceding operations to lax $\mathbb{E}_2$ operations in $\text{Fil}(\text{Syn})$, again by the Day-convolution construction. The resulting functor is precisely the displayed synthetic Tate filtration, as defined in [AP, Definition 6.10]. $\square$

4.2. Compatibility with realization. Write $\text{Re}: \text{Syn} \rightarrow \text{Sp}$ for the functor taking the colimit of the synthetic filtration. It is strong symmetric monoidal, since colimit is strong symmetric monoidal for Day convolution and $\text{Re}(\mathbb{S}_{\text{syn}}) \simeq \mathbb{S}$. The canonical identification $\text{Re}(\mathbf{T}_{\text{ev}}) \simeq \mathbb{S}[S^1]$ respects the bialgebra structures, by the ring-power construction in (8).

Proposition 4.5. *With the structure of Theorem 4.3, the natural equivalence*

$$\text{Re}(S_{\text{ev}}^*) \simeq S^{-*\mathbb{C}}$$

is an equivalence of $\mathbb{E}_2$ -algebras in $\text{Fil}(\text{Mod}_{\mathbb{S}[S^1]}(\text{Sp}))$ for diagonal tensor products. Moreover, there is a natural comparison

$$\text{Re}(F_T^*(X^{t\mathbf{T}_{\text{ev}}})) \longrightarrow F_T^*((\text{Re } X)^{tS^1}) \quad (11)$$

of lax $\mathbb{E}_2$ -monoidal functors. Realization on the left is applied in the synthetic direction, and the right side is the ordinary cellular Tate filtration with the structure of [AP, Theorems 3.4 and 3.6].

Proof. For an algebra–module pair (B, M) , the colimit formula [HRW, Remark A.1.5(3)] identifies

$$\operatorname{colim} \operatorname{fil}_{\mathbb{E}_\infty\text{-ev}, B}^*(M) \simeq \lim_{B \rightarrow C, \pi_* C \text{ even}} C \otimes_B M.$$

The base-change units define a canonical map from M to this limit. These maps are compatible with external tensor operations: after projection to any even algebra over a tensor product of base rings, both composites are the tensor product of the corresponding base-change units. The equality is an equality of the natural cones, so also supplies the higher coherences. The same cones show compatibility with restriction comparisons.

For the ring powers and sphere modules used above, this canonical map is an equivalence. Indeed, after taking colimits, the comparison in Lemma 2.3 is exactly this map, under the exhaustivity identification $\operatorname{colim} \operatorname{fil}_{\operatorname{pev}, B}^*(M) \simeq M$. Proposition 2.4 shows that the comparison is an equivalence in these cases. Its inverse therefore identifies the realizations of the external operations and restriction comparisons with the original operations and restrictions. Formula (10) consequently realizes to the diagonal tensor operation on sphere modules. Applying this observation to the termwise Day construction proves the assertion for the filtered representation spheres and their transported $\mathbb{E}_2$ -algebra structure.

For any $\mathbf{T}_{\text{ev}}$ -module Y , the fixed-point adjunction has counit

$$\operatorname{triv}(Y^{h\mathbf{T}_{\text{ev}}}) \longrightarrow Y.$$

Realization commutes with trivial actions because it respects the augmentations of $\mathbf{T}_{\text{ev}}$ and $\mathbb{S}[S^1]$. Realizing the counit and taking its adjoint therefore gives a natural map

$$\operatorname{Re}(Y^{h\mathbf{T}_{\text{ev}}}) \longrightarrow (\operatorname{Re} Y)^{hS^1}.$$

This comparison is lax symmetric monoidal. Indeed, the tensor operations on either fixed-point functor are adjoint to the tensor products of its counits. Since realization is strong symmetric monoidal, it carries these tensor products of counits, with their unit and higher coherences, to the corresponding tensor products after realization. Their adjoints are precisely the required compatibility diagrams.

Apply this map to $Y = S_{\text{ev}}^q \otimes_{\mathbb{S}_{\text{syn}}} X$ in each degree q . The coefficient equivalence just proved gives

$$\operatorname{Re}(S_{\text{ev}}^q \otimes_{\mathbb{S}_{\text{syn}}} X) \simeq S^{-q\mathbb{C}} \otimes \operatorname{Re} X.$$

Its compatibility with the $\mathbb{E}_2$ operations and the lax symmetric monoidal fixed-point comparison prove (11). $\square$

Corollary 4.6. *Let k be a commutative ring, regarded as an Eilenberg–Mac Lane $\mathbb{E}_\infty$ -ring, and equip $\tau_{\geq 2*}k \in \operatorname{Syn}$ with trivial $\mathbf{T}_{\text{ev}}$ -action. For X with compatible $\mathbf{T}_{\text{ev}}$ - and $\tau_{\geq 2*}k$ -module structures, the map (11) is a map in $\operatorname{Fil}(\operatorname{Mod}_k(\operatorname{Sp}))$. For commutative algebra objects X , it respects the $\mathbb{E}_2$ structures constructed above.*

Proof. The canonical map $\mathbb{S}_{\text{syn}} \rightarrow \tau_{\geq 2*}k$ realizes to $\mathbb{S} \rightarrow k$, and $\operatorname{Re}(\tau_{\geq 2*}k) \simeq k$. Thus realization induces a strong symmetric monoidal functor from $\operatorname{Mod}_{\tau_{\geq 2*}k}(\operatorname{Syn})$ to $\operatorname{Mod}_k(\operatorname{Sp})$. The tensor products with the coefficient spheres and the fixed-point counits in the preceding proof are linear over these scalar rings. The same adjunction argument in their module categories proves the assertion. $\square$

4.3. Larger domains for diagonal monoidality. Proposition 2.1 requires only even MU -homology. To extend the local monoidal construction, we impose flatness so that the class of inputs is closed under tensor products.

Corollary 4.7. *For every integer $a \geq 0$, consider the full subcategory of $\text{Mod}_{\mathbb{S}[(S^1)^a]}(\text{Sp})$ consisting of modules M such that UM is bounded below and MU_*UM is a flat graded MU_* -module concentrated in even degrees. This subcategory contains the unit and is closed under diagonal tensor products. On it, the perfect-even module filtration admits a lax symmetric monoidal refinement with values in*

$$(\text{Mod}_{\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}} a}}(\text{Syn}), \otimes_{\Delta}).$$

The comparison with the HRW module filtration is a natural equivalence on this subcategory.

Proof. For two underlying spectra UM and UN satisfying the hypotheses, the equivalence

$$(MU \otimes UM) \otimes_{MU} (MU \otimes UN) \simeq MU \otimes UM \otimes UN$$

and the flat Künneth calculation give

$$MU_*(UM \otimes UN) \cong MU_*UM \otimes_{MU_*} MU_*UN.$$

Here higher Tor groups vanish by flatness; this is the calculation used in [Pst, proof of Proposition 4.20]. Since MU is connective and both MU -modules are bounded below, the Künneth spectral sequence converges. The result is again flat and even, and $UM \otimes UN$ is bounded below. The unit has MU -homology MU_* , so the asserted closure properties follow.

For this enlarged subcategory, Corollary 2.5 supplies the restriction equivalences for every diagonal

$$\mathbb{S}[(S^1)^a] \longrightarrow \mathbb{S}[(S^1)^{an}] \quad (n \geq 0).$$

The ring tensor comparisons are still those of (8). Compose the external HRW operations with the inverses of these restriction comparisons, as in (10). External tensor products remain in the specified subcategory, and restriction leaves the underlying spectrum unchanged. Thus every cartesian restriction and every external product used in the coherence argument of Proposition 4.1 is available here. That argument gives the lax symmetric monoidal structure, including the nullary operation. Proposition 2.4 and Lemma 2.3 identify the perfect-even and HRW module functors; transport along this natural equivalence proves the claim for the perfect-even filtration. $\square$

Example 4.8. The preceding corollary applies to arbitrary torus actions on spectra whose underlying spectra are finite even-cell spectra or retracts of such spectra. Indeed, smashing a finite even-cell filtration with MU gives successive quotients that are even suspensions of MU . The resulting sequences on homotopy are short exact, with free quotient, and split as graded MU_* -modules. Induction gives finite free even MU -homology; retracts give projective even MU -homology. More generally, among finite spectra the condition of having projective even MU -homology is exactly the perfect-even condition [Pst, Proposition 3.17]. For example, finite complex projective spaces give coefficients $\Sigma_+^\infty \mathbb{C}P^n$ of this form.

Remark 4.9. The class of all spectra with even MU -homology is not closed under tensor products. For a prime p , the Moore spectrum $\mathbb{S}/p$ has MU -homology MU_*/p , concentrated in even degrees. The cofibre sequence

$$MU/p \xrightarrow{p} MU/p \longrightarrow MU \otimes (\mathbb{S}/p) \otimes (\mathbb{S}/p)$$

gives

$$MU_1((\mathbb{S}/p) \otimes (\mathbb{S}/p)) \cong \ker(p: \mathbb{F}_p \rightarrow \mathbb{F}_p) \cong \mathbb{F}_p.$$

Thus Proposition 2.1 applies to a larger class than the tensor-closed class in Corollary 4.7.

4.4. Invertibility of the coefficient multiplication.

Lemma 4.10. *For the specified coefficient algebra of Theorem 4.3, the unit and every multiplication map*

$$\mathbb{S}_{\text{syn}} \longrightarrow S_{\text{ev}}^0, \quad \mu_{a,q}: S_{\text{ev}}^a \otimes_{\Delta} S_{\text{ev}}^q \longrightarrow S_{\text{ev}}^{a+q} \quad (a, q \in \mathbb{Z})$$

are equivalences.

Proof. In the proof of [AP, Theorem 3.6], the ordinary coefficient algebra $S^{-*\mathbb{C}}$ is obtained by localizing the nonnegative coefficient algebra at

$$\alpha: \text{ins}^1(S^{-\mathbb{C}}) \longrightarrow R.$$

Here ins^1 denotes the free decreasing filtration generated in degree one. After localization, multiplication by α is an equivalence. The degree-zero term is the sphere, and its unit is an equivalence: on underlying π_0 , the unit law makes the scalar defining the unit a unit of $\mathbb{Z}$. The component of the localized action with target degree one is $S^{-\mathbb{C}} \otimes S^0 \rightarrow S^{-\mathbb{C}}$. Composing with the degree-zero unit identifies it with the degree-one component of α , which is therefore an equivalence. The localized action then identifies

$$S^{-\mathbb{C}} \otimes S^{-q\mathbb{C}} \longrightarrow S^{-(q+1)\mathbb{C}}$$

with an equivalence for every q . Iterating this action proves the same assertion for $a \geq 0$. For $a < 0$, associate the three factors of degrees $-a, a, q$: the products with first degree $-a$ and the product with first degree zero are already equivalences. Tensoring with $S^{a\mathbb{C}}$ is invertible, so two-out-of-three proves the assertion in this case as well. The strong local functor of Proposition 4.1 carries these specified operations to equivalences. Transport through the canonical perfect-even/HRW comparison, as in Theorem 4.3, gives the result for S_{ev}^* . $\square$

5. COORDINATEWISE TATE MULTIFILTRATIONS FOR TORUS ACTIONS

Fix $d \geq 1$. For $\mathbf{T}_{\text{ev}}$ -modules $Y_1, \dots, Y_d$, write $Y_1 \boxtimes \dots \boxtimes Y_d$ for their tensor product over $\mathbb{S}_{\text{syn}}$ with the separate action of each factor of $\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}}} d}$. For $X \in \text{Mod}_{\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}}} d}(\text{Syn})}$ and $\mathbf{q} = (q_1, \dots, q_d) \in \mathbb{Z}^d$, define

$$F_{T^d}^{\mathbf{q}}(X) := ((S_{\text{ev}}^{q_1} \boxtimes \dots \boxtimes S_{\text{ev}}^{q_d}) \otimes_{\Delta} X)^{h\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}}} d}}. \quad (12)$$

We call the diagram (12), indexed by $(\mathbb{Z}^{\text{op}})^d$, the coordinatewise synthetic Tate multifiltration.

Proposition 5.1. *The functor (12) is lax $\mathbb{E}_2$ -monoidal with values in $\text{Fun}((\mathbb{Z}^{\text{op}})^d, \text{Syn})$, equipped with Day convolution for coordinatewise addition. It is complete in each coordinate separately. If U forgets the torus action, its underlying multigraded synthetic spectrum is naturally*

$$\text{gr}^{\mathbf{q}} F_{T^d}(X) \simeq UX[-2(q_1 + \dots + q_d)](-q_1 - \dots - q_d), \quad (13)$$

where $\text{gr}^{\mathbf{q}}$ takes the associated graded in all d directions. Moreover, realization gives a natural comparison of lax $\mathbb{E}_2$ -monoidal multifiltered functors whose term at $\mathbf{q}$ is

$$\text{Re } F_{T^d}^{\mathbf{q}}(X) \longrightarrow \left((S^{-q_1\mathbb{C}} \boxtimes \dots \boxtimes S^{-q_d\mathbb{C}}) \otimes \text{Re } X \right)^{h(S^1)^d}. \quad (14)$$

On the right, $\boxtimes$ is the external smash product with its coordinate circle actions.

Proof. The external product of the d filtered coefficient algebras of Theorem 4.3 is an $\mathbb{E}_2$ -algebra in the indicated multifiltered category of $\mathbf{T}_{\text{ev}}^{\otimes_{\mathbb{S}_{\text{syn}}} d}$ -modules. Its operations apply the same little-disks operation in each coordinate and reorder tensor factors using the symmetry. Day convolution assembles these maps, with their full operadic coherences, just as in Lemma 4.2. Tensoring with this coefficient algebra is lax $\mathbb{E}_2$ -monoidal. Fixed points are lax symmetric monoidal, being right adjoint to the strong symmetric monoidal trivial-action functor. Their composite proves the first assertion.

The coefficient computation in [AP, proof of Proposition 6.14] gives $\mathrm{gr}^q S_{\mathrm{ev}}^* \simeq \mathbf{T}_{\mathrm{ev}}^\vee[-2q](-q)$ as $\mathbf{T}_{\mathrm{ev}}$ -modules. Tensor products and fixed points are exact, so taking the d cofibres in (12) replaces its coefficient by

$$(\mathbf{T}_{\mathrm{ev}}^\vee \boxtimes \cdots \boxtimes \mathbf{T}_{\mathrm{ev}}^\vee)[-2(q_1 + \cdots + q_d)](-q_1 - \cdots - q_d).$$

The coinduction identity used in that same proof gives $(\mathbf{T}_{\mathrm{ev}}^\vee \otimes_\Delta Y)^{h\mathbf{T}_{\mathrm{ev}}} \simeq UY$. Apply it one circle at a time, retaining the commuting actions of the remaining factors; the evaluation and adjunction maps defining this identity respect those actions. Iterated fixed points compute fixed points for the product, by their adjunctions. This proves (13).

To prove coordinatewise completeness, fix all but one coordinate. By the natural $\mathbb{S}_{\mathrm{syn}}$ -linear comparison in Proposition 2.1, the underlying transition $S_{\mathrm{ev}}^{q+1} \rightarrow S_{\mathrm{ev}}^q$ is the image of $S^{-2q-2} \rightarrow S^{-2q}$, whose class in $\pi_{-2}\mathbb{S}$ is zero. The perfect-even filtration preserves the zero object, so this transition is null as a map of $\mathbb{S}_{\mathrm{syn}}$ -modules. It remains null after tensoring with the other coefficients and with X . Such a countable tower has zero limit: its limit is the fibre of $1 - \text{shift}$ on the product, and the shift is nullhomotopic. The forgetful functor on modules creates limits and fixed points preserve limits, giving completeness in the chosen coordinate.

Finally, Proposition 4.5 identifies the realization of each coefficient algebra with the ordinary representation-sphere algebra. External tensor products preserve these identifications. The fixed-point comparison obtained by realizing the trivial-action counit is lax symmetric monoidal, by the same tensor-counit argument as in that proposition. This gives (14) with its stated $\mathbb{E}_2$ compatibility. $\square$

6. MULTIPLICATIVE CONNECTIVE COVERS

Let $\mathcal{C}$ be a presentably symmetric monoidal stable category whose tensor product preserves colimits in each variable. Suppose its t -structure is compatible with the tensor product and is right separated, as in Sp , $D(k)$, and the neutral synthetic category used here. We use the Beilinson t -structure on $\mathrm{Fil}(\mathcal{C})$, with conventions [Ant24, Definitions 3.24–3.25 and Remark 3.26]

$$\begin{aligned} F \in \mathrm{Fil}(\mathcal{C})_{\geq a}^{\mathrm{B}} &\iff \mathrm{gr}^q F \in \mathcal{C}_{\geq a-q} \text{ for every } q, \\ F \in \mathrm{Fil}(\mathcal{C})_{\leq a-1}^{\mathrm{B}} &\iff F^q \in \mathcal{C}_{\leq a-1-q} \text{ for every } q. \end{aligned}$$

The second condition is a condition on filtration terms. It cannot in general be replaced by an associated-graded condition without completeness.

Lemma 6.1. *Associated graded and doubling are symmetric monoidal for Day convolution. The Whitehead tower and the Beilinson Whitehead tower are lax symmetric monoidal. Completion is a symmetric monoidal localization when its target is equipped with completed Day convolution.*

Proof. For associated graded and the two Whitehead towers these are [Ant24, Proposition 8.4 and Lemmas 8.5, 8.7]. Doubling is left Kan extension along the additive functor $q \mapsto 2q$, so the Day convolution universal property identifies its tensor comparison with an equivalence. For completion, its kernel consists of constant filtrations. Tensoring a constant filtration with F gives a constant filtration with value its tensor product with $|F|$. The kernel is therefore a tensor ideal, which gives the asserted localization and the product $F \widehat{\otimes} G = \widehat{F} \otimes \widehat{G}$; see also [Ant24, Section 8]. $\square$

Lemma 6.2. *Let $\mathcal{O}$ be an ∞ -operad, let $Z \in \mathrm{Alg}_{\mathcal{O}}(\mathcal{C})$, and suppose*

$$Y^* \longrightarrow \mathrm{const}(Z)$$

is a morphism in $\mathrm{Alg}_{\mathcal{O}}(\mathrm{Fil}(\mathcal{C}))$. If each map $Y^a \rightarrow Z$ exhibits Y^a as $\tau_{\geq a}Z$, then there is a unique equivalence over $\mathrm{const}(Z)$

$$Y^* \simeq \tau_{\geq *}Z$$

in $\text{Alg}_{\mathcal{O}}(\text{Fil}(C))$, with a contractible space of choices. The same assertion applies to the Beilinson t -structure.

Proof. The truncation adjunction first identifies the underlying tower over Z . For input degrees $a_1, \dots, a_n$, put $a = \sum_i a_i$. Compatibility of the tensor product with the t -structure gives $\bigotimes_i Y^{a_i} \in \mathcal{C}_{\geq a}$. Thus

$$\text{Map}_C\left(\bigotimes_i Y^{a_i}, \tau_{\geq a} Z\right) \xrightarrow{\simeq} \text{Map}_C\left(\bigotimes_i Y^{a_i}, Z\right).$$

This is an equivalence of spaces, natural in the inputs. Each operadic operation into Z therefore lifts uniquely to the indicated term of the tower. The equivalence of mapping spaces also lifts every homotopy among composites, and hence the full operadic diagram. For the nullary operation one uses $\mathbf{1} \in \mathcal{C}_{\geq 0}$. The resulting equivalence of operadic towers is unique over Z . The Beilinson case uses its tensor compatibility from [lemma 6.1](#). $\square$

Décalage is defined by

$$(\text{Dec } F)^a = |\tau_{\geq a}^B F|.$$

It is lax symmetric monoidal: the Beilinson tower is lax symmetric monoidal and realization is symmetric monoidal. We record the multiplicative form of the doubling identity used in the HKR comparison.

Proposition 6.3. *For $Y \in \text{Alg}_{\mathbb{E}_2}(\text{Fil}(\text{Sp}))$, the equivalence of [\[AP, Proposition 4.6\]](#) refines naturally to*

$$\text{Dec}(\text{Dec}(d_! Y)) \simeq d_! \text{Dec}(Y)$$

as filtered $\mathbb{E}_2$ -algebras. The bifiltered refinement of [\[AP, Remark 4.7\]](#) is also multiplicative. These statements hold in $D(k)$ with its standard t -structure.

Proof. By [\[AP, Lemma 4.5\]](#), the map

$$\text{Dec}(d_! \tau_{\geq a}^B Y) \longrightarrow \text{Dec}(d_! Y)$$

exhibits its source as the Beilinson $2a$ -connective cover. As a varies, these maps form an operadic tower: apply the Beilinson tower, $d_!$, and décalage, using [lemma 6.1](#). The mapping-space argument of [lemma 6.2](#), with degrees $2a$, makes the connective-cover identification multiplicative. Realizing the original direction and using the evenness in [\[AP, Proposition 4.6\]](#) gives the displayed equivalence.

For the bifiltered statement, use the truncated filtration

$$(F^s Y)^t = Y^{\max(s, t)}.$$

There are compatible products $F^s Y \otimes F^{s'} Y \rightarrow F^{s+s'} Y$: the inequality

$$\max(s, t) + \max(s', t') \geq \max(s + s', t + t')$$

supplies the necessary transition map after multiplication in Y . The preceding connective-cover argument therefore applies as an operadic diagram in both parameters. It gives the refinement in [\[AP, Remark 4.7\]](#), including the specialization at the zero term used in its HKR comparison. Truncation, doubling, realization, and décalage are detected on underlying spectra. The connective-cover lifting argument takes place internally in $D(k)$, with tensor products over k , so the same proof applies there. $\square$

Lemma 6.4. *The forgetful functor from $\mathbb{E}_2$ -algebras in a complete filtered or complete multifiltered category creates sifted colimits. In particular, sifted left Kan extension of an $\mathbb{E}_2$ -algebra-valued functor has the prescribed underlying left Kan extension.*

Proof. By [lemma 6.1](#), these are presentably symmetric monoidal categories with tensor product preserving colimits separately. The assertion is the sifted-colimit theorem for operadic algebras [\[HA, Proposition 3.2.3.1\]](#). $\square$

7. MULTIPLICATIVE COMPARISON OF HKR FILTRATIONS

Let k be a classical commutative ring, and let A be an animated k -algebra. We write $D(k)$ for the derived ∞ -category of k -modules and use homological shifts: a discrete module $M[n]$ has its homotopy in degree n . Hochschild homology has its canonical circle action, and

$$\mathrm{HC}^-(A/k) = \mathrm{HH}(A/k)^{hS^1}, \quad \mathrm{HP}(A/k) = \mathrm{HH}(A/k)^{tS^1}.$$

We use the notation of [AP, Recollections 7.3, 7.6, and 7.8] for the Antieau, Bhatt–Lurie, and Raksit filtrations. All tensor products of complete filtered objects in this section are completed Day products. The notation $\mathrm{BiFil}^c(D(k))$ means completeness in each of the two filtration directions.

7.1. The filtered constructions. Raksit’s filtered Hochschild homology

$$F_{\mathrm{Rak}}^\bullet \mathrm{HH}(A/k) = \mathrm{HH}^{\mathrm{fl}}(A/k)$$

is a filtered derived commutative k -algebra with a filtered circle action [Rak, Definition 6.2.5]. We regard this entire filtered object as a module over the filtered circle. Applying the k -linear synthetic Tate construction gives its HKR filtration on periodic cyclic homology; applying the synthetic Tate *filtration* retains an additional index:

$$F_T^\bullet F_{\mathrm{Rak}}^\bullet \mathrm{HP}(A/k).$$

This is the construction of [AP, Notation 7.11]. The multiplicative construction and its k -linear realization compatibility in Proposition 4.5 and Corollary 4.6 equip it with an $\mathbb{E}_2$ -algebra structure. It is complete in both directions by [AP, Lemma 7.12].

For a finitely generated polynomial k -algebra P , the Bhatt–Lurie bifiltration is the Beilinson connective-cover tower

$$F_{\mathrm{BhLu}}^r F_T^\bullet \mathrm{HP}(P/k) = \tau_{\geq r}^B(F_T^\bullet \mathrm{HP}(P/k)). \quad (15)$$

Here the Beilinson t -structure is taken in the Tate direction. The ordinary cellular Tate filtration is lax $\mathbb{E}_2$ -monoidal [AP, Theorem 3.4]; its Beilinson tower is therefore a bifiltered $\mathbb{E}_2$ -algebra by Lemma 6.1. For arbitrary animated algebras, the construction is extended from polynomial algebras by sifted left Kan extension, with completion in the Tate direction [AP, Recollection 7.6].

We use these constructions to specify the algebra structures under comparison. Left Kan extensions can be formed in $\mathbb{E}_2$ -algebras: the forgetful functor from $\mathbb{E}_2$ -algebras creates sifted colimits [HA, §3.2.3]. Completion is a symmetric monoidal localization, so this also applies in the completed filtered categories. In the Bhatt–Lurie construction the extension agrees with extension into bicomplete objects, as in [BL, Remark 6.3.5] and [AP, proof of Proposition 7.14].

7.2. The Bhatt–Lurie–Raksit comparison. The comparison is determined by two assertions: on polynomial algebras, the canonical realization map exhibits the Raksit tower as a Beilinson connective-cover tower; on arbitrary animated algebras, both constructions are sifted left Kan extensions. The first assertion identifies the multiplication by connective-cover rigidity, and the second extends this identification.

Theorem 7.1. *There is a natural equivalence of bifiltered $\mathbb{E}_2$ -algebras*

$$F_{\mathrm{BhLu}}^\bullet F_T^\bullet \mathrm{HP}(A/k) \simeq \sigma^*(F_T^\bullet F_{\mathrm{Rak}}^\bullet \mathrm{HP}(A/k)) \quad \text{in } \mathrm{Alg}_{\mathbb{E}_2}(\mathrm{BiFil}^c(D(k))),$$

where $\sigma(r, s) = (s, r)$ exchanges the two indexing directions.

Proof. Let P be a finitely generated polynomial k -algebra. The canonical map

$$|F_T^\bullet F_{\mathrm{Rak}}^\bullet \mathrm{HP}(P/k)|_{\mathrm{Rak}} \longrightarrow F_T^\bullet \mathrm{HP}(P/k) \quad (16)$$

of [AP, Notation 7.11] is an $\mathbb{E}_2$ -algebra map by Proposition 4.5 and Corollary 4.6. The subscript on realization specifies that the Raksit direction is realized and the Tate direction

is retained. The realization adjunction consequently gives a morphism of $\mathbb{E}_2$ -algebras in the category of towers of Tate-filtered objects, with target the constant tower on $F_T^\bullet \mathrm{HP}(P/k)$.

By [AP, Lemma 7.13], its stage with Raksit index r is a Beilinson r -connective cover of $F_T^\bullet \mathrm{HP}(P/k)$. Thus Lemma 6.2 identifies the tower, including its $\mathbb{E}_2$ -structure and its map to the constant tower, with (15). Exchanging the coordinates is symmetric monoidal for Day convolution. This proves the assertion on polynomial algebras.

The underlying Raksit–Tate bifiltration preserves sifted colimits in $\mathrm{BiFil}^c(D(k))$ by [AP, proof of Proposition 7.14]. Its $\mathbb{E}_2$ -algebra-valued version therefore preserves those colimits as well, since they are created on underlying objects. The polynomial equivalence extends to the claimed equivalence on all animated k -algebras by the defining left Kan extension of the Bhatt–Lurie construction. $\square$

7.3. The Antieau–Bhatt–Lurie comparison. We recall the additional filtration used in Antieau’s construction. For a polynomial k -algebra P , set

$$F_H^r F_T^s \mathrm{HP}(P/k) = F_T^s((\tau_{\geq r} \mathrm{HH}(P/k))^{tS^1}).$$

The index r is the homotopy-filtration index. Taking the double-speed Beilinson connective-cover tower in this direction produces

$$F_{BH}^a F_H^r F_T^s \mathrm{HP}(P/k) = \left(\tau_{\geq 2a}^{B,H} (F_H^\bullet F_T^\bullet \mathrm{HP}(P/k)) \right)^{r,s}.$$

The superscript B, H specifies the filtration used for the Beilinson t -structure. This is extended from polynomial algebras with the completions specified in [AP, Recollection 7.3]; the Antieau bifiltration is its homotopy-filtration-zero part $F_{BH}^\bullet F_H^0 F_T^\bullet \mathrm{HP}(A/k)$. The ordinary Whitehead tower, the cellular Tate filtration, the Beilinson tower, and evaluation at filtration zero give this construction an $\mathbb{E}_2$ -algebra structure. Evaluation at zero is lax monoidal by the Day-convolution structure maps.

Proposition 7.2. *For every animated k -algebra A , the comparison of [AP, Lemma 7.10] refines to an equivalence*

$$F_{\mathrm{Ant}}^\bullet F_T^\bullet \mathrm{HP}(A/k) \simeq F_{\mathrm{BhLu}}^\bullet F_T^\bullet \mathrm{HP}(A/k)$$

of bifiltered $\mathbb{E}_2$ -algebras.

Proof. For an $\mathbb{E}_2$ -algebra X with circle action, the natural maps

$$d_! F_T^\bullet((\tau_{\geq r} X)^{tS^1}) \longrightarrow d_! F_T^\bullet(X^{tS^1})$$

assemble into a morphism of bifiltered $\mathbb{E}_2$ -algebras, with constant target in the r -direction. Indeed, the Whitehead tower is lax symmetric monoidal, the ordinary cellular Tate filtration is lax $\mathbb{E}_2$ -monoidal, and $d_!$ is symmetric monoidal. By [AP, Theorem 5.1(1)], these maps are Beilinson r -connective covers. Lemma 6.2 therefore identifies this tower with the Beilinson tower of the target as a bifiltered $\mathbb{E}_2$ -algebra.

Apply this assertion to $X = \mathrm{HH}(P/k)$ for P polynomial. Apply the double-speed Beilinson tower in the homotopy direction, and then evaluate that direction at zero. These operations preserve the $\mathbb{E}_2$ -structure. The bifiltered double-décalage identity of Proposition 6.3, in the form of [AP, Remark 4.7], identifies the resulting tower with $\tau_{\geq \bullet}^B(F_T^\bullet \mathrm{HP}(P/k))$. This is precisely the polynomial comparison in [AP, proof of Lemma 7.10], now as a comparison of $\mathbb{E}_2$ -algebras.

For general A , the same proof of [AP, Lemma 7.10] passes among completion in the Tate and homotopy directions, completion in all three directions, and completion in the Tate and Beilinson-homotopy directions. The identifications of the underlying Kan extensions are [Ant24, Lemmas 4.8 and 4.9] and [BL, Remark 6.3.5]. Carry out these Kan extensions in $\mathbb{E}_2$ -algebras. Their underlying objects are the indicated Kan extensions, and the completion functors are symmetric monoidal localizations. The polynomial equivalence hence extends through the same canonical comparison maps and gives the asserted equivalence. $\square$

Corollary 7.3. *The three HKR/Tate bifiltrations admit natural equivalences*

$$F_{\text{Ant}}^\bullet F_T^\bullet \text{HP}(A/k) \simeq F_{\text{BhLu}}^\bullet F_T^\bullet \text{HP}(A/k) \simeq \sigma^*(F_T^\bullet F_{\text{Rak}}^\bullet \text{HP}(A/k))$$

in $\text{Alg}_{\mathbb{E}_2}(\text{BiFil}^c(D(k)))$. Their Tate-filtration-zero parts give equivalent filtered $\mathbb{E}_2$ -algebras on $\text{HC}^-(A/k)$.

Proof. Combine Theorem 7.1 and Proposition 7.2. Taking the Tate-filtration-zero part is lax monoidal and gives HC^- , as in [AP, Theorem 7.9]. $\square$

7.4. The double associated graded. The comparison has a direct spectral-sequence consequence. Apply the spectral-sequence construction in either of the corresponding filtration directions, retaining or taking associated graded in the other direction. Corollary 7.3 identifies the resulting spectral sequences, after exchanging coordinates, including their products and differentials. Indeed, an equivalence of filtered $\mathbb{E}_2$ -algebras identifies the multiplicative exact couples. Thus a product or differential calculated using one of the three constructions can be transported to the other two.

Let $L_{A/k}$ be the cotangent complex, and let $\Lambda_A^q L_{A/k}$ denote its derived exterior power, with value zero for $q < 0$. Raksit's associated-graded theorem gives

$$\text{gr}_{\text{Rak}}^q \text{HH}(A/k) \simeq (\Lambda_A^q L_{A/k})[q]$$

[Rak, Theorem 6.2.6 and Remark 6.2.10]. Combining this with [AP, Proposition 6.14] gives a natural equivalence of underlying bigraded objects

$$\text{gr}_T^s \text{gr}_{\text{Rak}}^r \text{HP}(A/k) \simeq (\text{gr}_{\text{Rak}}^{r+s} \text{HH}(A/k))[-2s] \simeq (\Lambda_A^{r+s} L_{A/k})[r-s]. \quad (17)$$

In particular, the Hochschild shift $[r+s]$ is retained before the Tate shift $[-2s]$ is applied. When $s = 0$ and A is smooth, this places $\Omega_{A/k}^r$ in homological degree r .

7.5. The exterior-power algebra structure. The double associated graded admits a more explicit description than (17). Give the exterior-power summand $(\Lambda_A^m L_{A/k})[m]$ bidegree $(m, 0)$, and give a Laurent variable u bidegree $(-1, 1)$ and homological degree -2 . The resulting bigraded algebra has component

$$((\Lambda_A^{r+s} L_{A/k})[r+s])u^s = (\Lambda_A^{r+s} L_{A/k})[r-s].$$

Its algebra structure is specified as follows. For a polynomial k -algebra P , use the wedge product on differential forms and the ordinary Laurent product in u . For animated algebras, take the sifted left Kan extension of this functor in bigraded $\mathbb{E}_\infty$ -algebras. Its underlying components are the displayed derived exterior powers; equivalently, this is the multiplicative derived exterior-power construction of [Rak, Theorems 5.3.6 and 6.2.6], with the indicated regrading.

Theorem 7.4. *For every animated commutative k -algebra A , the equivalence (17) refines naturally to an equivalence of bigraded $\mathbb{E}_2$ -algebras*

$$\text{gr}_T^* \text{gr}_{\text{Rak}}^* \text{HP}(A/k) \simeq \left(\bigoplus_{m \geq 0} (\Lambda_A^m L_{A/k})[m] \right) [u^{\pm 1}]$$

with the algebra structure and bidegrees specified above. In particular, the specified $\mathbb{E}_2$ structure on the double associated graded admits this natural $\mathbb{E}_\infty$ refinement. The same description applies to the Antieau and Bhatt–Lurie bifiltrations through Corollary 7.3.

For an abelian group G , let G_{disc} be the discrete category with object set G . Write $\text{Gr}_G(D(k)) = \text{Fun}(G_{\text{disc}}, D(k))$ for the ∞ -category of G -graded objects, with degree-preserving maps and Day tensor product

$$(X \otimes_k^{\mathbf{L}} Y)^g = \bigoplus_{a+b=g} X^a \otimes_k^{\mathbf{L}} Y^b.$$

The following lemma will be applied to the double associated graded for polynomial algebras, after its binary products have been computed.

Lemma 7.5. *Let k be a classical commutative ring and $h: G \rightarrow \mathbb{Z}$ an additive map. For $n = 2$ or $n = \infty$, consider the full ∞ -category of unital $\mathbb{E}_n$ -algebras in $\mathrm{Gr}_G(D(k))$ whose underlying objects satisfy*

$$\pi_i Z^g = 0 \quad \text{for } i \neq h(g).$$

The functor $Z \mapsto \{\pi_{h(g)} Z^g\}_{g \in G}$ identifies this ∞ -category with the nerve of the ordinary category of unital G -graded k -algebras with associative products $M_g \otimes_k M_{g'} \rightarrow M_{g+g'}$ satisfying

$$xy = (-1)^{h(g)h(g')}yx \quad (x \in M_g, y \in M_{g'}).$$

Morphisms in the latter category are degree-preserving algebra homomorphisms. Under these identifications, restriction from $\mathbb{E}_\infty$ to $\mathbb{E}_2$ corresponds to the identity functor. In particular, for a specified such $\mathbb{E}_2$ -algebra Z , the space of $\mathbb{E}_\infty$ -algebras B equipped with an equivalence $B|_{\mathbb{E}_2} \simeq Z$ is contractible.

Proof. Multimorphism spaces. Let $X_1, \dots, X_m, Y$ be graded objects satisfying the displayed purity condition. Fix $g_1 + \dots + g_m = g$ and put $d = h(g) = \sum_i h(g_i)$. The derived tensor product

$$C = X_1^{g_1} \otimes_k^{\mathbf{L}} \dots \otimes_k^{\mathbf{L}} X_m^{g_m} \quad \text{belongs to } D(k)_{\geq d},$$

whereas Y^g is concentrated in degree d . For $j > 0$, the orthogonality of the t -structure gives

$$\pi_j \mathrm{Map}_{D(k)}(C, Y^g) = \pi_0 \mathrm{Map}_{D(k)}(C[j], Y^g) = 0,$$

since $C[j] \in D(k)_{\geq d+j}$ and $Y^g \in D(k)_{\leq d}$. Truncating C at d therefore identifies this mapping space with the discrete set

$$\mathrm{Hom}_k \left(\bigotimes_{i=1}^m \pi_{h(g_i)} X_i^{g_i}, \pi_{h(g)} Y^g \right). \quad (18)$$

The tensor product in (18) is underived: higher Tor groups contribute to homotopy of C above degree d . The nullary case has source k in grading zero and is covered by $h(0) = 0$.

A graded multimorphism consists of these maps for all input degrees, so its space is a product of the discrete spaces (18). This remains discrete for infinite G . The identifications respect composition and units, since they are induced by homotopy in the lowest possible degree. Interchanging adjacent inputs gives the Koszul sign $(-1)^{h(g_i)h(g_{i+1})}$. Consequently, the colored operad of pure objects and their multimorphisms identifies with the ordinary multicategory of G -graded k -modules with this symmetry.

Algebra structures and refinements. In every arity, the operation spaces of both $\mathbb{E}_2$ and $\mathbb{E}_\infty$ are connected. A map from either operad to an operad with discrete multimorphism spaces is determined by its map on connected components, including the permutation actions and the substitution laws. The operad of connected components is the commutative operad in both cases. Thus either algebra category identifies with the ordinary category of commutative algebra objects for the indicated symmetry. Because the calculation included different objects X_i and Y , it also identifies all algebra morphisms: they are precisely the degree-preserving module maps commuting with products and units, and their mapping spaces are discrete.

These identifications commute with restriction of operads and with passage to underlying objects. In particular, after fixing an underlying graded object Z , its algebra-structure space is the discrete set of the specified products on $\{\pi_{h(g)} Z^g\}_g$. Restriction from $\mathbb{E}_\infty$ to $\mathbb{E}_2$ induces an equivalence of the full algebra categories above. Taking the homotopy fiber on their maximal subgroupoids over a specified $\mathbb{E}_2$ -algebra proves the contractibility assertion. For the ungraded discrete-algebra version of this argument, see [HA, Proposition 7.1.3.18]. $\square$

Proof of Theorem 7.4. First let $A = P$ be polynomial. We calculate using the Bhatt–Lurie tower and transport across Theorem 7.1. We first track the degree-zero multiplication through the Serre construction. For right Day convolution on increasing diagrams, the functor $B \mapsto \text{fib}(B_0 \rightarrow B_1)$ is strong symmetric monoidal. Indeed, the tensor product of B and C has terms

$$B_0 \otimes C_0, \quad (B_1 \otimes C_0) \times_{B_1 \otimes C_1} (B_0 \otimes C_1)$$

in degrees zero and one. Exactness of tensor products identifies the fibre between these terms with $\text{fib}(B_0 \rightarrow B_1) \otimes \text{fib}(C_0 \rightarrow C_1)$. The analogous total-fibre calculation for tensor cubes supplies all arities and their coherences; the right Day unit has terms $\mathbb{S}, 0, 0, \dots$

The Serre formula of [AP, Remark 2.5] gives

$$\text{fib}(\text{Serre}(X)_0 \rightarrow \text{Serre}(X)_1) \simeq \text{fib}(X_\infty \rightarrow \text{cofib}(X_0 \rightarrow X_\infty)) \simeq X_0.$$

This is a monoidal natural equivalence for the structure of [AP, Proposition 2.12]. One may check this on the linear Yoneda generators used in that proposition: both sides vanish on positive-degree generators, take the degree-zero generator to the unit, and their nonzero tensor comparisons are the unit comparison. Both functors preserve colimits, so the Day universal property proves the assertion. Here D denotes monoidal duality on dualizable spectra. Dualizing now identifies the zeroth associated graded of the dual Serre construction with DX_0 , including its algebra structure.

For the cellular construction in [AP, proof of Theorem 3.6], $X_0 = \Sigma_+^\infty S^1$ with its cartesian diagonal coalgebra. Thus the degree-zero coefficient is the pointwise function algebra $D(\Sigma_+^\infty S^1)$. The subsequent localization preserves the nonnegative pieces and this identification, as in that proof. The coinduction equivalence, normalized by evaluation at the identity of the circle, identifies

$$(D(\Sigma_+^\infty S^1) \otimes_\Delta Y)^{hS^1} \simeq UY$$

and its binary products with the underlying products of Y .

It remains to normalize the other degrees. The same construction localizes at the degree-one representation-sphere generator; its associated-graded action is invertible. After extension of scalars to k , the oriented complex line identifies this representation coefficient with the trivial circle module $k[-2]$. The trivialization is unique after fixing its underlying orientation: the automorphism space of $k[-2]$ as a k -module is the discrete group $k^\times$, and BS^1 is simply connected. These trivializations therefore respect binary tensor products. Combining the invertible generator with the preceding coinduction identification gives precisely the Laurent binary products $u^p u^q = u^{p+q}$ and the original products on UY .

For polynomial algebras, the Bhatt–Lurie construction is the Beilinson connective-cover tower (15). Taking its associated graded in the HKR direction extracts

$$(\pi_{r-s}(\text{HH}(P/k)[-2s]))[r-s] = \Omega_{P/k}^{r+s}[r-s].$$

The products are the induced products on homotopy. The multiplicative HKR theorem identifies these with wedge products [Rak, Theorem 6.2.6 and Remark 6.2.10]. Together with the preceding Laurent generator, this is exactly the product in the statement. Every component is concentrated in degree $r-s$. Apply Lemma 7.5 with $G = \mathbb{Z}^2$ and $h(r, s) = r-s$. The computed product identifies the two ordinary bigraded algebras in that lemma, hence gives the claimed $\mathbb{E}_2$ equivalence and an $\mathbb{E}_\infty$ refinement on polynomial algebras. The assertion about morphisms makes this comparison natural in P .

For a general animated A , we extend this polynomial comparison by sifted left Kan extension. The bifiltration preserves sifted colimits in bicomplete objects, as used in Theorem 7.1. Taking the double associated graded preserves these colimits, since completion does not change associated graded. Sifted colimits of operadic algebras are created on underlying objects. Thus the polynomial equivalence extends to animated A by sifted

left Kan extension. On the right this is exactly the algebra structure specified before the theorem, and its components are the derived exterior powers. This proves the result. $\square$

For example, if $A = k[x_1, \dots, x_n]$, the common double associated graded is

$$A \otimes_k \Lambda_k(dx_1, \dots, dx_n) \otimes_k k[u^{\pm 1}],$$

where dx_i has bidegree $(1, 0)$ and homological degree 1, and u has bidegree $(-1, 1)$ and homological degree -2 .

8. DÉCALAGE OF THE SYNTHETIC TATE FILTRATION

We use the neutral t -structure on Syn and on $\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$. A synthetic spectrum is neutral connective precisely when each of its internal filtration terms is connective. The t -structure is compatible with the tensor product, and the forgetful functor from $\mathbf{T}_{\text{ev}}$ -modules is t -exact [AR, Construction 2.2, Example 2.77, and Remark 2.78]. An internal filtration shift (m) is neutral t -exact.

To identify a Beilinson connective cover, we need connectivity of its source and coconnectivity of its cofiber. The graded formula of [AP, Proposition 6.14] gives the first assertion directly. For the second we use completeness, then recover the required bounds on filtration terms from finite quotients.

Lemma 8.1. *For every $\mathbf{T}_{\text{ev}}$ -module X , the external filtration $F_T^*(X^{t\mathbf{T}_{\text{ev}}})$ is complete.*

Proof. This is the case $d = 1$ of Proposition 5.1. $\square$

Lemma 8.2. *Suppose $F \in \text{Fil}(\mathcal{C})$ is complete and $\text{gr}^q F \in \mathcal{C}_{\leq a-1-q}$ for every q . Then $F^q \in \mathcal{C}_{\leq a-1-q}$ for every q , and consequently $F \in \text{Fil}(\mathcal{C})_{\leq a-1}^{\text{B}}$.*

Proof. For $m > q$, the quotient $\text{cofib}(F^m \rightarrow F^q)$ is a finite extension of $\text{gr}^q F, \dots, \text{gr}^{m-1} F$. Each lies in $\mathcal{C}_{\leq a-1-q}$, so the quotient does as well. Completeness and exactness of limits give

$$F^q \simeq \lim_{m \rightarrow +\infty} \text{cofib}(F^m \rightarrow F^q).$$

The coconnective subcategory is closed under limits. This proves the bound, which is precisely the Beilinson coconnectivity condition. $\square$

Theorem 8.3. *For every $X \in \text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$ and $a \in \mathbb{Z}$, the map*

$$d_! F_T^*((\tau_{\geq a}^{\text{N}} X)^{t\mathbf{T}_{\text{ev}}}) \longrightarrow d_! F_T^*(X^{t\mathbf{T}_{\text{ev}}})$$

is a Beilinson a -connective cover for the neutral t -structure.

Proof. The graded formula of [AP, Proposition 6.14] and the definition of doubling give, naturally in a $\mathbf{T}_{\text{ev}}$ -module Y ,

$$\text{gr}^q(d_! F_T^*(Y^{t\mathbf{T}_{\text{ev}}})) \simeq \begin{cases} Y[-q](-q/2), & q \text{ even}, \\ 0, & q \text{ odd}. \end{cases} \quad (19)$$

For $Y = \tau_{\geq a}^{\text{N}} X$, the right side is neutral $(a - q)$ -connective. Thus the source is Beilinson a -connective.

Each Tate-filtration term is an exact functor of X : tensoring is exact and fixed points, being a right adjoint between stable categories, preserve finite limits and hence finite colimits. The cofiber of the displayed map is therefore

$$d_! F_T^*((\tau_{< a}^{\text{N}} X)^{t\mathbf{T}_{\text{ev}}}).$$

It is complete by lemma 8.1; doubling preserves completeness. Its graded pieces in equation (19) lie in $\text{Syn}_{\leq a-1-q}^{\text{N}}$. By lemma 8.2, the cofiber is Beilinson $(a - 1)$ -coconnective. These are the two conditions defining the asserted connective cover. $\square$

Proof of theorem 1.3. Realize the Beilinson tower identified in theorem 8.3 in the original Tate direction. Doubling does not change this colimit, and [AP, Proposition 6.16] identifies the colimit of the synthetic Tate filtration with the synthetic Tate construction. Hence, for every a ,

$$\begin{aligned} (\mathrm{Dec}_N(d_! F_T^*(X^{t\mathbf{T}_{\mathrm{ev}}}))^a &\simeq \left| d_! F_T^*((\tau_{\geq a}^N X)^{t\mathbf{T}_{\mathrm{ev}}}) \right| \\ &\simeq (\tau_{\geq a}^N X)^{t\mathbf{T}_{\mathrm{ev}}}. \end{aligned}$$

Naturality makes this an equivalence of filtered objects.

The neutral Whitehead tower is lax symmetric monoidal by lemma 6.1. The synthetic Tate construction is lax symmetric monoidal by [AR, Lemma 2.60]. Their composite, using the termwise Day operations, supplies a lax symmetric monoidal refinement of the right side. Transport across the preceding equivalence proves the second assertion. $\square$

Remark 8.4. For an $\mathbb{E}_2$ -algebra X , theorem 1.1 and lemma 6.2 identify the tower in theorem 8.3 as an operadic tower. Applying décalage gives a filtered $\mathbb{E}_2$ -algebra from the synthetic Tate filtration.

9. COEFFICIENTS AND RELATIVE TENSOR PRODUCTS

The multiplication of Theorem 4.4 supplies a coefficient algebra

$$\mathcal{R}_T = F_T^*(\mathbb{S}_{\mathrm{syn}}^{t\mathbf{T}_{\mathrm{ev}}}) \in \mathrm{Alg}_{\mathbb{E}_2}(\mathrm{Fil}(\mathrm{Syn})),$$

where $\mathbb{S}_{\mathrm{syn}}$ has trivial $\mathbf{T}_{\mathrm{ev}}$ -action. We prove a completed relative tensor-product formula for arbitrary inputs.

Fix the associative structure underlying the specified $\mathbb{E}_2$ structure. The operations with inputs $(\mathbb{S}_{\mathrm{syn}}, X)$ and $(X, \mathbb{S}_{\mathrm{syn}})$ give commuting left and right $\mathcal{R}_T$ -actions on $F_T^*(X^{t\mathbf{T}_{\mathrm{ev}}})$. These actions make each value an $\mathcal{R}_T$ -bimodule, and the relative tensor product below uses the right action on its first factor and the left action on its second. The category of bimodules is monoidal under this tensor product [HA, Proposition 4.4.3.12].

Lax associativity gives a natural bimodule comparison

$$\kappa_{X,Y}: F_T^*(X^{t\mathbf{T}_{\mathrm{ev}}}) \otimes_{\mathcal{R}_T} F_T^*(Y^{t\mathbf{T}_{\mathrm{ev}}}) \longrightarrow F_T^*((X \otimes_{\Delta} Y)^{t\mathbf{T}_{\mathrm{ev}}}). \quad (20)$$

Indeed, the iterated operations give a compatible cone from the two-sided bar construction, whose term in simplicial degree m is

$$F_T^*(X^{t\mathbf{T}_{\mathrm{ev}}}) \otimes \mathcal{R}_T^{\otimes m} \otimes F_T^*(Y^{t\mathbf{T}_{\mathrm{ev}}}).$$

Realizing this cone in $\mathrm{Fil}(\mathrm{Syn})$ gives (20) [HA, Theorem 4.4.2.8].

Theorem 9.1. *For all $X, Y \in \mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn})$, the completed relative comparison is an equivalence of $\mathcal{R}_T$ -bimodules:*

$$F_T^*(X^{t\mathbf{T}_{\mathrm{ev}}}) \widehat{\otimes}_{\mathcal{R}_T} F_T^*(Y^{t\mathbf{T}_{\mathrm{ev}}}) \xrightarrow{\sim} F_T^*((X \otimes_{\Delta} Y)^{t\mathbf{T}_{\mathrm{ev}}}).$$

The uncompleted comparison is already an equivalence when either input is a compact object of $\mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn})$. It is also an equivalence when either input has the form $S_{\mathrm{ev}}^a \otimes_{\Delta} \mathrm{triv}(V)$ with V dualizable in Syn .

The proof first identifies tensoring with a coefficient sphere as external filtration translation. The Euler cofiber then places the free even-circle module in the class where the relative comparison is an equivalence. Completeness lets us pass from these free modules to arbitrary inputs. We record the two auxiliary facts needed for this last step and for synthetic coefficients.

Lemma 9.2. *The synthetic Tate filtration, viewed as a functor*

$$\mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn}) \longrightarrow \mathrm{Fil}^c(\mathrm{Syn}),$$

preserves small colimits. Its lift to $\mathcal{R}_T$ -bimodules in this complete filtered category also preserves small colimits.

Proof. Its values are complete by Lemma 8.1. Colimits in $\mathrm{Fil}^c(\mathrm{Syn})$ are computed by first taking the colimit in $\mathrm{Fil}(\mathrm{Syn})$ and then completing. Completion preserves associated graded, so associated graded preserves these colimits. For a diagram $\{X_i\}$, apply associated graded to the natural map

$$\mathrm{colim}_i^{\mathrm{Fil}^c(\mathrm{Syn})} F_T^*(X_i^{t\mathbf{T}_{\mathrm{ev}}}) \longrightarrow F_T^*((\mathrm{colim}_i X_i)^{t\mathbf{T}_{\mathrm{ev}}}).$$

By [AP, Proposition 6.14], its degree- q component becomes

$$\mathrm{colim}_i U X_i[-2q](-q) \longrightarrow U(\mathrm{colim}_i X_i)[-2q](-q),$$

where $U: \mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn}) \rightarrow \mathrm{Syn}$ is forgetful. This is an equivalence because U preserves colimits. Associated graded detects equivalences between complete filtrations: a complete filtration with zero associated graded is constant and hence zero. The colimit comparison is therefore an equivalence.

The tensor product in the complete filtered category preserves colimits separately. Its bimodule forgetful functor consequently creates colimits [HA, Proposition 4.3.3.9], so the bimodule lift has the same property. $\square$

Proposition 9.3. *Let V be dualizable in Syn , and equip it with trivial $\mathbf{T}_{\mathrm{ev}}$ -action. For every $Y \in \mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn})$, there is a natural equivalence of $\mathcal{R}_T$ -bimodules*

$$F_T^*((\mathrm{triv}(V) \otimes_{\Delta} Y)^{t\mathbf{T}_{\mathrm{ev}}}) \simeq \mathrm{ins}^0(V) \otimes F_T^*(Y^{t\mathbf{T}_{\mathrm{ev}}}).$$

In particular, $F_T^(\mathrm{triv}(V)^{t\mathbf{T}_{\mathrm{ev}}}) \simeq \mathrm{ins}^0(V) \otimes \mathcal{R}_T$.*

Proof. Trivial action is strong symmetric monoidal. Moving V across mapping spaces using its dual and then applying the fixed-point adjunction gives

$$V \otimes Z^{h\mathbf{T}_{\mathrm{ev}}} \simeq (\mathrm{triv}(V) \otimes_{\Delta} Z)^{h\mathbf{T}_{\mathrm{ev}}}.$$

Apply this with $Z = S_{\mathrm{ev}}^q \otimes_{\Delta} Y$ in every degree q . Tensoring a filtration with $\mathrm{ins}^0(V)$ gives this termwise tensor product. The projection formula is adjoint to the tensor product of V with the fixed-point counit. It therefore respects the coefficient operations, which act on the S_{ev}^q factors, and hence both $\mathcal{R}_T$ -actions. $\square$

9.1. Translation and the Euler cofiber. For any filtration Z and integer a , Day convolution gives

$$(\mathrm{ins}^{-a}(\mathbb{S}_{\mathrm{syn}}) \otimes Z)^q \simeq Z^{q+a}. \quad (21)$$

This translation acts on the external Tate index q .

Proposition 9.4. *For $a \in \mathbb{Z}$ and $X \in \mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn})$, there is a natural equivalence of underlying filtered synthetic spectra*

$$F_T^*((S_{\mathrm{ev}}^a \otimes_{\Delta} X)^{t\mathbf{T}_{\mathrm{ev}}}) \simeq \{F_T^{q+a}(X^{t\mathbf{T}_{\mathrm{ev}}})\}_{q \in \mathbb{Z}}.$$

The uncompleted relative comparison (20) is an equivalence whenever either input is S_{ev}^a and the other input is arbitrary.

Proof. In degree q , move S_{ev}^a to the left of S_{ev}^q and use $\mu_{a,q}$. Applying fixed points gives an equivalence

$$b_X^a(q): (S_{\mathrm{ev}}^q \otimes_{\Delta} S_{\mathrm{ev}}^a \otimes_{\Delta} X)^{h\mathbf{T}_{\mathrm{ev}}} \xrightarrow{\sim} (S_{\mathrm{ev}}^{a+q} \otimes_{\Delta} X)^{h\mathbf{T}_{\mathrm{ev}}}. \quad (22)$$

The multiplication is compatible with the filtration transitions, so these are equivalences of filtered objects.

When $X = \mathbb{S}_{\mathrm{syn}}$, this choice is right $\mathcal{R}_T$ -linear and gives

$$F_T^*((S_{\mathrm{ev}}^a)^{t\mathbf{T}_{\mathrm{ev}}}) \simeq \mathrm{ins}^{-a}(\mathbb{S}_{\mathrm{syn}}) \otimes \mathcal{R}_T \quad \text{as right } \mathcal{R}_T\text{-modules.} \quad (23)$$

Indeed, after applying the fixed-point counits, the two composites in the right-linearity diagram both have inputs ordered as $S_{\mathrm{ev}}^a, S_{\mathrm{ev}}^p, S_{\mathrm{ev}}^q$. Their equality is the associativity identity

$$\mu_{a,p+q}(\mathrm{id} \otimes \mu_{p,q}) \simeq \mu_{a+p,q}(\mu_{a,p} \otimes \mathrm{id}).$$

The associative structure supplies these identities with all the coherences needed for the right-module diagram.

Use (23) to identify the first variable of $\kappa_{S_{\text{ev}}, Y}^a$, and use b_Y^a to identify its target. The same associativity calculation identifies its defining bar cone with the external translation of the usual action cone. Its realization is therefore the unit-action equivalence

$$(\text{ins}^{-a}(\mathbb{S}_{\text{syn}}) \otimes \mathcal{R}_T) \otimes_{\mathcal{R}_T} F_T^*(Y^{t\mathbf{T}_{\text{ev}}}) \xrightarrow{\sim} \text{ins}^{-a}(\mathbb{S}_{\text{syn}}) \otimes F_T^*(Y^{t\mathbf{T}_{\text{ev}}}).$$

This argument identifies the comparison itself, including the whole bar cone, with an equivalence. The original comparison is a bimodule map, so an equivalence on underlying objects is an equivalence of bimodules. For the other variable, use the mirror construction, multiplying S_{ev}^a on the right; it gives the required left-module identification. $\square$

Proof of Theorem 9.1. Fix Y . The class of X for which the uncompleted comparison is an equivalence is closed under finite stable constructions and retracts, because both sides are exact functors of X : tensor products and fixed points are exact, and relative tensor products preserve colimits in each variable. It contains every S_{ev}^a by Proposition 9.4. It is also closed under tensoring with a trivial-action dualizable synthetic spectrum. To see this, apply Proposition 9.3 on both sides of the comparison and move the dualizable coefficient through the relative bar construction. The resulting comparison is the tensor product of that coefficient with $\kappa_{X, Y}$. The compatibility of these maps follows from their construction using the same coefficient multiplication and fixed-point counits.

The proof of [AP, Proposition 6.14] gives a cofiber sequence of $\mathbf{T}_{\text{ev}}$ -modules

$$S_{\text{ev}}^1 \longrightarrow S_{\text{ev}}^0 \longrightarrow \mathbf{T}_{\text{ev}}^\vee.$$

Thus $\mathbf{T}_{\text{ev}}^\vee$ belongs to this class. By the equivariant duality

$$\mathbf{T}_{\text{ev}}^\vee \simeq \mathbf{T}_{\text{ev}}-1$$

of [AP, Recollection 6.6], it follows that $\mathbf{T}_{\text{ev}} \otimes_{\mathbb{S}_{\text{syn}}} \mathbb{S}_{\text{syn}}(n)$ belongs to the class for every $n \in \mathbb{Z}$: the required suspensions and internal shifts are trivial-action invertible synthetic coefficients.

These free modules are compact generators of $\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$. The free filtration spectra $\text{ins}^n(\mathbb{S})$ compactly generate $\text{Fil}(\text{Sp})$ by evaluation in degree n . The free-module adjunctions, first for $\mathbb{S}_{\text{syn}}$ and then for $\mathbf{T}_{\text{ev}}$, give the indicated compact generators. Their thick closure is the subcategory of compact $\mathbf{T}_{\text{ev}}$ -modules. This proves the assertions about the uncompleted comparison in the first variable; the same proof applies to the second.

Now consider the completed comparison for the same fixed Y . Both sides preserve all small colimits in X , by Lemma 9.2, the symmetric monoidal completion of Lemma 6.1, and separate colimit preservation of relative tensor products. Its equivalence locus is therefore a localizing subcategory. It contains the compact generators just exhibited, because their uncompleted comparisons are already equivalences to a complete target. It is consequently the whole category, proving the displayed equivalence for all X, Y . $\square$

Corollary 9.5. *Let V be a finite-dimensional virtual complex representation of S^1 , and let $n \in \mathbb{Z}$. There are natural equivalences*

$$\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^{nV}) \simeq (\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^V))^{\otimes_{\Delta} n}.$$

Negative tensor powers use the inverse $\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^{-V})$, and the zeroth power is the trivial-action unit $\mathbb{S}_{\text{syn}}$.

Moreover, $F_T^((\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^V))^{t\mathbf{T}_{\text{ev}}})$ is invertible under completed relative tensor product of $\mathcal{R}_T$ -bimodules, with inverse $F_T^*((\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^{-V}))^{t\mathbf{T}_{\text{ev}}})$. For every $n \in \mathbb{Z}$,*

$$F_T^*((\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^{nV}))^{t\mathbf{T}_{\text{ev}}}) \simeq \left(F_T^*((\text{fil}_{\text{pev}, \mathbb{S}[S^1]}(S^V))^{t\mathbf{T}_{\text{ev}}}) \right)^{\widehat{\otimes}_{\mathcal{R}_T} n}.$$

Here negative powers use this bimodule inverse and the zeroth power is $\mathcal{R}_T$.

Proof. The underlying spectrum of a virtual complex representation sphere has even dimension. Its tensor inverse is the sphere of the negative virtual representation, and $S^V \otimes_{\Delta} S^W \simeq S^{V+W}$. Transport the strong local functor of Proposition 4.1 through the canonical comparison of Proposition 2.4 and Lemma 2.3. It preserves these products, their units, and their inverses, proving the first assertion.

Apply Theorem 9.1 to these tensor products. For $W = -V$ it gives the unit bimodule $\mathcal{R}_T$ in either order, establishing the asserted inverse. Iteration gives the formula for all integer powers, with the associativity and unit coherences inherited from the relative comparisons. $\square$

10. RECONSTRUCTION OF MODULE CATEGORIES

In this section set

$$\mathcal{C} = \text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn}), \quad \mathcal{D} = \text{Fil}^c(\text{Syn}), \quad F(X) = F_T^*(X^{t\mathbf{T}_{\text{ev}}}).$$

The tensor products on $\mathcal{C}$ and $\mathcal{D}$ are the diagonal tensor product and completed Day convolution, respectively. Write RMod for right modules. The specified associative structure gives a lift

$$F^r: \mathcal{C} \longrightarrow \text{RMod}_{\mathcal{R}_T}(\mathcal{D})$$

by retaining the right coefficient action on $F(X)$.

Theorem 10.1. *The functor F^r is an equivalence of presentable stable ∞ -categories.*

Proof. Lemma 9.2 shows that F^r preserves small colimits. We first identify its mapping comparison on coefficient spheres with internal shifts, then use the Euler cofiber to obtain full faithfulness. The same spheres supply free generators of the target.

For a dualizable $V \in \text{Syn}$, equations (23) and Proposition 9.3 give

$$F^r(S_{\text{ev}}^a \otimes_{\Delta} \text{triv}(V)) \simeq \text{ins}^{-a}(V) \hat{\otimes} \mathcal{R}_T. \quad (24)$$

Step objects are complete, and the step–evaluation adjunction restricts to $\mathcal{D}$. Thus the free right-module adjunction identifies maps from (24) to $F^r(X)$ with

$$\begin{aligned} \text{map}_{\text{Syn}}(V, F^{-a}(X)) &\simeq \text{map}_{\mathcal{C}}(\text{triv}(V), S_{\text{ev}}^{-a} \otimes_{\Delta} X) \\ &\simeq \text{map}_{\mathcal{C}}(S_{\text{ev}}^a \otimes_{\Delta} \text{triv}(V), X). \end{aligned} \quad (25)$$

The last equivalence uses the invertibility of the coefficient spheres. We check the normalization so that these adjunctions identify the actual map induced by F^r .

For $V = \mathbb{S}_{\text{syn}}$, the free generator of (24) corresponds to a map

$$e_a: \text{ins}^{-a}(\mathbb{S}_{\text{syn}}) \longrightarrow F(S_{\text{ev}}^a).$$

Under the fixed-point adjunction, its value in degree $-a$ is the inverse of the equivalence

$$S_{\text{ev}}^{-a} \otimes_{\Delta} S_{\text{ev}}^a \xrightarrow{\text{swap}} S_{\text{ev}}^a \otimes_{\Delta} S_{\text{ev}}^{-a} \xrightarrow{\mu_{a,-a}} S_{\text{ev}}^0 \simeq \mathbb{S}_{\text{syn}}. \quad (26)$$

Indeed, the right-module translation (22) carries e_a to the translated unit of $\mathcal{R}_T$, and its degree- $-a$ formula is precisely (26). Let η_a denote this inverse. For $f: S_{\text{ev}}^a \rightarrow X$, restriction of $F(f)$ along e_a is adjoint to

$$\mathbb{S}_{\text{syn}} \xrightarrow{\eta_a} S_{\text{ev}}^{-a} \otimes_{\Delta} S_{\text{ev}}^a \xrightarrow{\text{id} \otimes f} S_{\text{ev}}^{-a} \otimes_{\Delta} X.$$

Tensoring with S_{ev}^{-a} is an equivalence, and precomposition with η_a is an equivalence on mapping spectra. Thus the displayed operation is an equivalence on maps. Tensoring this diagram with $\text{triv}(V)$ gives the same conclusion for $S_{\text{ev}}^a \otimes_{\Delta} \text{triv}(V)$: the projection formula is the mate of the fixed-point counit tensored with V , so it preserves this generator and its naturality. Consequently the equivalence (25) is inverse to the mapping comparison induced by F^r .

Fix $X \in \mathcal{C}$. The objects P for which

$$\mathrm{map}_{\mathcal{C}}(P, X) \longrightarrow \mathrm{map}_{\mathrm{RMod}_{\mathcal{R}_T}(\mathcal{D})}(F^r(P), F^r(X))$$

is an equivalence form a localizing subcategory: F^r preserves colimits, and mapping out of colimits gives limits on both sides. The preceding calculation places every $S_{\mathrm{ev}}^a \otimes_{\Delta} \mathrm{triv}(\mathbb{S}_{\mathrm{syn}}(n))$ in this subcategory. The Euler cofiber and equivariant duality

$$S_{\mathrm{ev}}^1 \longrightarrow S_{\mathrm{ev}}^0 \longrightarrow \mathbf{T}_{\mathrm{ev}}^{\vee}, \quad \mathbf{T}_{\mathrm{ev}}^{\vee} \simeq \mathbf{T}_{\mathrm{ev}}-1,$$

then place every $\mathbf{T}_{\mathrm{ev}} \otimes_{\mathbb{S}_{\mathrm{syn}}} \mathbb{S}_{\mathrm{syn}}(n)$ there. These free modules generate $\mathcal{C}$, as in the proof of Theorem 9.1. Hence the mapping comparison is an equivalence for every P and X , proving full faithfulness.

The objects $\mathrm{ins}^a(\mathbb{S}_{\mathrm{syn}}(n))$, for $a, n \in \mathbb{Z}$, generate $\mathrm{Fil}(\mathrm{Syn})$ under stable colimits. Completion is a colimit-preserving localization and fixes these step objects, so they also generate $\mathcal{D}$. The free-module bar resolution now shows that

$$\mathrm{ins}^a(\mathbb{S}_{\mathrm{syn}}(n)) \widehat{\otimes} \mathcal{R}_T$$

generate $\mathrm{RMod}_{\mathcal{R}_T}(\mathcal{D})$ under stable colimits. Every such object is the image of $S_{\mathrm{ev}}^{-a} \otimes_{\Delta} \mathrm{triv}(\mathbb{S}_{\mathrm{syn}}(n))$ by (24). A fully faithful colimit-preserving functor has a colimit-closed essential image, so F^r is essentially surjective. $\square$

Theorem 10.2. *Let $B \in \mathrm{Alg}_{\mathbb{E}_1}(\mathcal{C})$. The specified multiplication and action maps induce an equivalence of presentable stable ∞ -categories*

$$F_B: \mathrm{RMod}_B(\mathcal{C}) \xrightarrow{\sim} \mathrm{RMod}_{F(B)}(\mathcal{D}), \quad M \longmapsto F(M).$$

It fits into a commutative diagram

$$\begin{array}{ccc} \mathrm{RMod}_B(\mathcal{C}) & \xrightarrow[\sim]{F_B} & \mathrm{RMod}_{F(B)}(\mathcal{D}) \\ \downarrow & & \downarrow \mathrm{Res}_{\mathcal{R}_T} \\ \mathcal{C} & \xrightarrow[\sim]{F^r} & \mathrm{RMod}_{\mathcal{R}_T}(\mathcal{D}), \end{array}$$

where restriction uses the unital algebra map $\mathcal{R}_T \rightarrow F(B)$. In particular, F_B preserves mapping spectra, composition, and compact objects.

Proof. The forgetful functor from right B -modules is monadic, with free-module monad

$$X \longmapsto X \otimes_{\Delta} B.$$

Likewise, extension and restriction of scalars give an adjunction

$$-\widehat{\otimes}_{\mathcal{R}_T} F(B): \mathrm{RMod}_{\mathcal{R}_T}(\mathcal{D}) \rightleftarrows \mathrm{RMod}_{F(B)}(\mathcal{D}) : \mathrm{Res}_{\mathcal{R}_T}.$$

Restriction is conservative and preserves geometric realizations, which are computed on underlying objects. The adjunction is therefore monadic by [HA, Theorem 4.7.3.5]; its monad sends P to $P \widehat{\otimes}_{\mathcal{R}_T} F(B)$, with the resulting right $\mathcal{R}_T$ -action. Its unit comes from $\mathcal{R}_T \rightarrow F(B)$, and its multiplication comes from the balanced product of $F(B)$.

Theorem 9.1 identifies these monads across F^r . Explicitly, it supplies natural right-module equivalences

$$F^r(X) \widehat{\otimes}_{\mathcal{R}_T} F(B) \xrightarrow{\sim} F^r(X \otimes_{\Delta} B). \quad (27)$$

The relative comparisons respect all ordered iterated products: their bar cones are induced by the same operations of the specified lax $\mathbb{E}_1$ structure. In particular, their two composites on three inputs agree by its associativity, and its unit operation gives the unit comparison. Iterating gives, for every $m \geq 0$,

$$F^r(X) \widehat{\otimes}_{\mathcal{R}_T} F(B) \widehat{\otimes}_{\mathcal{R}_T}^m \simeq F^r(X \otimes_{\Delta} B^{\otimes_{\Delta} m}),$$

compatibly with inserting units and multiplying adjacent copies of B , including their higher associativity coherences. Completion preserves the bar constructions and these comparison maps. Thus (27) is an equivalence of monads, not only of their underlying endofunctors.

Transporting algebras for these monads through Theorem 10.1 proves the assertion. The transported action is the composite

$$F(M) \hat{\otimes}_{\mathcal{R}_T} F(B) \xrightarrow{\sim} F(M \otimes_{\Delta} B) \longrightarrow F(M).$$

Precomposing with the map from the unbalanced tensor product recovers the action supplied by the original lax structure. The resulting equivalence is therefore F_B itself, and its underlying right coefficient module is $F^r(M)$, as asserted in the diagram. An equivalence of stable categories preserves mapping spectra and composition; an equivalence of presentable categories preserves colimits and hence preserves and reflects compactness. $\square$

Corollary 10.3. *Let $B \in \text{Alg}_{\mathbb{E}_1}(\mathcal{C})$, let M be a right B -module, and let N be a left B -module. There is a natural equivalence*

$$F(M) \hat{\otimes}_{F(B)} F(N) \xrightarrow{\sim} F(M \otimes_B N).$$

The equivalence is compatible with any further commuting module actions and with iterated relative tensor products.

Proof. Apply F to the two-sided bar construction for $M \otimes_B N$. It preserves its realization in $\mathcal{D}$ by Lemma 9.2. Iterated relative comparisons identify its degree- m term with

$$F(M) \hat{\otimes}_{\mathcal{R}_T} F(B) \hat{\otimes}_{\mathcal{R}_T}^m F(N).$$

The unit, action, and multiplication maps agree by the same coherences used in (27). Its realization is $F(M) \hat{\otimes}_{F(B)} F(N)$: this is the relative bar construction over the coefficient algebra, equivalently the ordinary relative tensor product after restriction of scalars [HA, Theorem 4.4.2.8]. The bar cones preserve outer actions, and the iterated bar constructions give the stated associativity compatibility. $\square$

11. FILTERED ALGEBRAS WITH DIFFERENT MODULE CATEGORIES

We construct two synthetic Tate algebras with equivalent underlying filtrations and equivalent algebras in Tate degree zero, but different categories of complete filtered modules. Their multiplication in nonzero Tate degrees determines this difference.

Put $k = \mathbb{F}_2$. We use the Postnikov t -structure on Syn of [AR, Construction 2.15]. Its heart is the category of graded $\mathbb{Z}[\eta]/(2\eta)$ -modules with the Koszul symmetry, where η has degree one. Let A be the commutative algebra in Syn corresponding to the heart algebra $k[\eta^{\pm 1}]$. More precisely, the functor π_0^P from connective objects to the heart is symmetric monoidal for the heart tensor product. Its right adjoint, the heart inclusion, is lax symmetric monoidal, and hence carries $k[\eta^{\pm 1}]$ to this commutative algebra A . The class η has homotopy degree one and internal filtration degree one. The underlying filtration of A is

$$A^r \simeq \Sigma^r Hk \quad (r \in \mathbb{Z}),$$

with zero transition maps. In particular, $\text{Re}(A) = 0$. Equip A with trivial $\mathbf{T}_{\text{ev}}$ -action and write

$$\mathcal{C}_A = \text{Mod}_{A \otimes_{\mathbb{S}_{\text{syn}}} \mathbf{T}_{\text{ev}}}(\text{Mod}_A(\text{Syn})).$$

Its tensor product is the diagonal tensor product relative to A .

Lemma 11.1. *There is a symmetric monoidal equivalence between $\mathcal{C}_A$ and the derived category of $\mathbb{Z}/2$ -graded k -modules, with parity convolution as tensor product. Let L correspond to k in odd parity and chain degree zero. Then*

$$L \otimes_A L \simeq A, \quad A \otimes_{\mathbb{S}_{\text{syn}}} S_{\text{ev}}^q \simeq L^{\otimes_A q}[-2q](-q) \quad (q \in \mathbb{Z}).$$

Negative tensor powers use $L^{-1} \simeq L$.

Proof. The free weight shifts $A(n)$ compactly generate $\text{Mod}_A(\text{Syn})$, by the free-module and evaluation adjunctions. Multiplication by η^n identifies An with A for every $n \in \mathbb{Z}$. Thus A is a compact generator, and its endomorphism spectrum is

$$\text{map}_A(A, A) \simeq A^0 \simeq Hk.$$

The symmetric monoidal compact-unit reconstruction [HA, Proposition 7.1.2.7] gives $\text{Mod}_A(\text{Syn}) \simeq D(k)$ as symmetric monoidal categories.

Set $H = A \otimes_{\mathbb{S}_{\text{syn}}} \mathbf{T}_{\text{ev}}$. The underlying-module splitting $\mathbf{T}_{\text{ev}} \simeq \mathbb{S}_{\text{syn}} \oplus \mathbb{S}_{\text{syn}}1$ places H in the Postnikov heart. Therefore [AR, Lemma 2.69] identifies its algebra structure as

$$H = A[d]/(d^2 - \eta d).$$

Its tensor powers are finite free A -modules in the heart, so the mapping spaces between them are discrete. Thus this identification includes the coherences of the Hopf structure. Here d has homotopy and internal degree one, and is primitive. The latter assertion follows already before inverting η : the counits determine the two linear terms of its coproduct, and a term proportional to $d \otimes d$ would require a coefficient of degree minus one in $\mathbb{Z}[\eta]/(2\eta)$. Consequently $e = \eta^{-1}d$ satisfies

$$e^2 = e, \quad \Delta(e) = e \otimes 1 + 1 \otimes e.$$

The two idempotent summands identify H with $A \times A$. Its modules split into the characters $e = 0$ and $e = 1$, and the coproduct adds these characters modulo two. This proves the claimed symmetric monoidal equivalence. In particular, L is the character on which d acts by η .

To identify the coefficient spheres, base change the Euler cofiber of [AP, proof of Proposition 6.14]:

$$A \otimes_{\mathbb{S}_{\text{syn}}} S_{\text{ev}}^1 \longrightarrow A \longrightarrow H^\vee.$$

Let e_0, e_1 be the basis dual to $1, d$. The dual action satisfies

$$de_0 = 0, \quad de_1 = e_0 + \eta e_1.$$

The map $A \rightarrow H^\vee$ is dual to the augmentation and sends 1 to e_0 . Its quotient is $L-1$, so its fiber is $L[-2](-1)$. Finally, the specified coefficient products are equivalences by Lemma 4.10. Their iterates and inverses give the formula for every q . $\square$

Using this parity description, define commutative algebra objects $B_0, B_1 \in \mathcal{C}_A$ by

$$B_0 = A \oplus Ax, \quad |x|_{\text{par}} = 1, \quad x^2 = 0, \quad B_1 = A \oplus Ax, \quad |x|_{\text{par}} = 1, \quad x^2 = 1.$$

Both have underlying equivariant object $A \oplus L$ and unit the inclusion of A . These are ordinary parity-graded commutative algebras in chain degree zero, transported through Lemma 11.1. Write

$$\mathcal{S}_i = F_T^*(B_i^{t\mathbf{T}_{\text{ev}}}) \quad (i = 0, 1),$$

with the specified associative structures underlying their $\mathbb{E}_2$ structures. For a presentable stable category, the superscript ω denotes its full subcategory of compact objects.

Theorem 11.2. *The underlying filtered synthetic spectra of $\mathcal{S}_0$ and $\mathcal{S}_1$ are canonically equivalent. Their terms are*

$$\mathcal{S}_i^q \simeq A[-2q](-q),$$

and their transition maps are zero. Their algebras in Tate degree zero are both A , and

$$\text{Re}(\mathcal{S}_i^q) = 0 \quad (q \in \mathbb{Z}), \quad \text{colim}_q \mathcal{S}_i^q = 0.$$

Their complete filtered module categories satisfy

$$\begin{aligned} K_0(\text{RMod}_{\mathcal{S}_0}(\text{Fil}^c(\text{Syn}))^\omega) &\cong \mathbb{Z}^2, \\ K_0(\text{RMod}_{\mathcal{S}_1}(\text{Fil}^c(\text{Syn}))^\omega) &\cong \mathbb{Z}. \end{aligned}$$

In particular, these two filtered algebras are not Morita equivalent.

Proof. Fixed points of an H -module select its even character summand. Lemma 11.1 therefore gives

$$F_T^q(A^{t\mathbf{T}_{\text{ev}}}) \simeq \begin{cases} A[-2q](-q), & q \text{ even}, \\ 0, & q \text{ odd}, \end{cases} \quad F_T^q(L^{t\mathbf{T}_{\text{ev}}}) \simeq \begin{cases} 0, & q \text{ even}, \\ A[-2q](-q), & q \text{ odd}. \end{cases}$$

Successive coefficient spheres have opposite characters. The mapping spectrum between their base changes is zero, so every coefficient transition, and hence every induced Tate transition, vanishes. Since B_0 and B_1 have the same underlying equivariant object, the equality of their entire underlying filtrations also follows directly from functoriality. Their degree-zero fixed points are their common even subalgebra A . A filtration with zero transitions has zero colimit. Each term is a unital A -module; since realization is symmetric monoidal and $\text{Re}(A) = 0$, its realization vanishes.

The differing products can be described directly. Put

$$\mathcal{R}_A = F_T^*(A^{t\mathbf{T}_{\text{ev}}}), \quad \mathcal{M} = F_T^*(L^{t\mathbf{T}_{\text{ev}}}).$$

The relative comparison of Corollary 10.3 identifies the specified map

$$\mathcal{M} \hat{\otimes}_{\mathcal{R}_A} \mathcal{M} \xrightarrow{\sim} F_T^*((L \otimes_A L)^{t\mathbf{T}_{\text{ev}}}) \simeq \mathcal{R}_A.$$

Both $\mathcal{S}_i$ have underlying coefficient bimodule $\mathcal{R}_A \oplus \mathcal{M}$. Their odd-odd products are, respectively, zero and this equivalence.

Apply Theorem 10.2 to the underlying associative algebra B_i in $\text{Mod}_{\mathbf{T}_{\text{ev}}}(\text{Syn})$. Its modules acquire their A -action by restriction along $A \rightarrow B_i$. As B_i is a commutative A -algebra, this identifies the source module category with $\text{RMod}_{B_i}(\mathcal{C}_A)$. We obtain equivalences

$$\text{RMod}_{B_i}(\mathcal{C}_A) \xrightarrow{\sim} \text{RMod}_{\mathcal{S}_i}(\text{Fil}^c(\text{Syn})). \quad (28)$$

On the target, restriction along the scalar map $\text{ins}^0(A) \rightarrow \mathcal{R}_A \rightarrow \mathcal{S}_i$ provides the corresponding A -action. By the completion adjunction, an action of the step algebra $\text{ins}^0(A)$ on a complete filtration is equivalently a termwise A -action; completeness is detected on the underlying filtration.

It remains to compute the source categories in (28). The free right modules on the two parity generators form the compact generator

$$P_i = B_i \oplus (L \otimes_A B_i) \quad \text{of } \text{RMod}_{B_i}(\mathcal{C}_A).$$

By the free-module adjunction, every matrix entry of $\text{map}_{B_i}(P_i, P_i)$ is a parity component of B_i . This endomorphism spectrum is concentrated in degree zero. Its two diagonal entries are k , and its two off-diagonal entries are generated by x . For $i = 0$, both composites of the off-diagonal generators vanish. The resulting algebra is the radical-square-zero quiver algebra

$$\Gamma = k \left(0 \begin{smallmatrix} \xrightarrow{a} \\ \xleftarrow{b} \end{smallmatrix} 1 \right) / (ab, ba).$$

For $i = 1$, the two composites are the respective diagonal identities, and the resulting algebra is $M_2(k)$. Derived Morita theory [HA, Theorem 7.1.2.1 and Remark 7.1.2.3] consequently gives

$$\text{RMod}_{B_0}(\mathcal{C}_A) \simeq D(\Gamma), \quad \text{RMod}_{B_1}(\mathcal{C}_A) \simeq D(M_2(k)) \simeq D(k).$$

These are equivalences of presentable stable categories, and therefore preserve compact objects. Perfect Γ -modules are represented by bounded complexes of finitely generated projectives. The two indecomposable projectives, corresponding to its primitive idempotents, therefore generate their Grothendieck group. Euler pairings with the two simple modules give the two coordinate functionals. Hence $K_0(\text{Perf}(\Gamma)) \cong \mathbb{Z}^2$, while $K_0(\text{Perf}(k)) \cong \mathbb{Z}$. Together with (28), this proves the asserted formulas. $\square$

After forgetting the circle action, even the A -algebras B_0 and B_1 are isomorphic: in characteristic two, the substitution $x \mapsto 1 + x$ interchanges the two presentations. This isomorphism mixes the two characters. The theorem thus detects the interaction between the circle action and the multiplication through the complete filtered module category.

12. AN OBSTRUCTION TO HIGHER REFINEMENTS

The coefficient construction has no $\mathbb{E}_3$ refinement. A related obstruction applies to synthetic Tate filtrations when their filtration-zero structure is required to agree with canonical fixed points.

Lemma 12.1. *Let G^* be an $\mathbb{E}_3$ -algebra in $\mathrm{Fil}(D(\mathbb{F}_2))$. For $q \geq 0$, the lower-indexed Dyer–Lashof operation at filtration zero carries the image of $\pi_m G^q$ into the image of $\pi_{2m+2} G^{2q}$.*

Proof. The binary filtered operad action has the form

$$\mathbb{E}_3(2)_+ \otimes_{\Sigma_2} (G^q \otimes_{\mathbb{F}_2} G^q) \longrightarrow G^{2q}.$$

The extended-power construction of Q_2 applied to a representative of a class in $\pi_m G^q$ thus produces a class in $\pi_{2m+2} G^{2q}$. Compatibility of the operad action with filtration transitions identifies its image at zero with Q_2 of the original image. Here Q_2 is the operation defined for $\mathbb{E}_3$ -algebras, with homotopy degree $m \mapsto 2m + 2$, as in [AP, Warning 3.5]. $\square$

Corollary 12.2. *The ordinary cellular Tate filtration of the trivial-action $\mathbb{F}_2$ admits no $\mathbb{E}_3$ -algebra structure in $\mathrm{Fil}(D(\mathbb{F}_2))$ whose filtration-zero structure is the canonical $\mathbb{E}_3$ -structure on $\mathbb{F}_2^{hS^1} = C^*(\mathbb{CP}^\infty; \mathbb{F}_2)$.*

Proof. For $q \geq 0$, this filtration is

$$F_T^q(\mathbb{F}_2^{tS^1}) \simeq \tau_{\leq -2q} C^*(\mathbb{CP}^\infty; \mathbb{F}_2),$$

with its canonical maps into cochains [AP, Warning 3.5]. The generator $t \in H^2(\mathbb{CP}^\infty; \mathbb{F}_2)$, regarded as a class in homotopy degree -2 , lifts to filtration one. But $\pi_{-2} F_T^2(\mathbb{F}_2^{tS^1}) = 0$. Lemma 12.1 would therefore imply $Q_2(t) = 0$. On canonical cochains, the same operation is $Q_2(t) = Q^0(t) = t$, again by [AP, Warning 3.5]. This is a contradiction. $\square$

Proposition 12.3. *The $\mathbb{E}_2$ -algebra S_{ev}^* of Theorem 4.3 admits no unital $\mathbb{E}_3$ refinement in $\mathrm{Fil}(\mathrm{Mod}_{\mathbf{T}_{\mathrm{ev}}}(\mathrm{Syn}))$.*

Proof. Such a refinement would realize to an $\mathbb{E}_3$ -algebra structure on $S^{-*\mathbb{C}}$, by the strong symmetric monoidality of realization and Proposition 4.5. Tensoring with the trivial-action $\mathbb{F}_2$ and taking ordinary circle fixed points would then give an $\mathbb{E}_3$ -algebra structure on its cellular Tate filtration.

At coefficient-filtration zero the unit map is an equivalence $\mathbb{S} \simeq S^0$. It identifies the resulting unital $\mathbb{E}_3$ -algebra with the monoidal unit, whose algebra structure is unique. Hence tensoring with $\mathbb{F}_2$ and applying the canonical fixed-point functor gives the canonical cochain $\mathbb{E}_3$ -structure at zero. This contradicts Corollary 12.2. $\square$

We next verify the realization comparison on a specific synthetic input. Let

$$\mathbb{F}_{2,\mathrm{fil}} := \tau_{\geq 2*} \mathbb{F}_2 = \mathrm{ins}^0(\mathbb{F}_2) \in \mathrm{Syn},$$

with trivial $\mathbf{T}_{\mathrm{ev}}$ -action and its canonical commutative algebra structure. Its module category supplies the scalar structure in the next statements.

Lemma 12.4. *The canonical comparison of Proposition 4.5 is an equivalence*

$$\mathrm{Re} F_T^*(\mathbb{F}_{2,\mathrm{fil}}^{t\mathbf{T}_{\mathrm{ev}}}) \xrightarrow{\sim} F_T^*(\mathbb{F}_2^{tS^1})$$

in $\mathrm{Alg}_{\mathbb{E}_2}(\mathrm{Fil}(D(\mathbb{F}_2)))$.

Proof. The filtered Hochschild homology of $\mathbb{F}_2$ over itself is $\mathbb{F}_{2,\text{fil}}$, with trivial filtered circle action. Apply [AP, Lemma 7.13] to this polynomial $\mathbb{F}_2$ -algebra on zero variables. It identifies the canonical internal-filtration- r row of its synthetic Tate filtration with the Beilinson r -connective cover of the ordinary cellular Tate filtration.

By [AP, Proposition 6.14], the associated graded in Tate degree q of that row is $\mathbb{F}_2[-2q]$ if $q \leq -r$, and zero otherwise. The map to the ordinary associated graded is an equivalence in the former range. The row is complete by [AP, Lemma 7.12]; therefore its tail for $q > -r$ is zero. Starting at this zero tail and descending in q , the cofiber sequences for the associated graded identify its term in degree q with

$$\tau_{\geq 2r} F_T^q(\mathbb{F}_2^{tS^1}) \longrightarrow F_T^q(\mathbb{F}_2^{tS^1}).$$

Indeed, the ordinary term has one copy of $\mathbb{F}_2$ in each even homotopy degree at most $-2q$, and the row retains exactly the degrees from $2r$ through $-2q$. This also identifies the canonical map, by induction through those cofiber sequences.

Realization is the colimit as $r \rightarrow -\infty$. The displayed Postnikov truncations exhaust the ordinary term, proving the comparison is an equivalence. Its $\mathbb{F}_2$ -linearity and $\mathbb{E}_2$ compatibility are Corollary 4.6 and Proposition 4.5. $\square$

Proposition 12.5. *The filtered $\mathbb{F}_{2,\text{fil}}$ -algebra $F_T^*(\mathbb{F}_{2,\text{fil}}^{t\mathbf{T}_{\text{ev}}})$ admits no $\mathbb{E}_3$ refinement for which the canonical filtration-zero realization comparison*

$$\text{Re}(\mathbb{F}_{2,\text{fil}}^{h\mathbf{T}_{\text{ev}}}) \longrightarrow \mathbb{F}_2^{hS^1}$$

is a map of $\mathbb{E}_3$ - $\mathbb{F}_2$ -algebras, where the target has its canonical cochain structure.

Proof. Realization induces a strong symmetric monoidal functor from $\mathbb{F}_{2,\text{fil}}$ -modules in Syn to $D(\mathbb{F}_2)$. A refinement as in the statement would thus give an $\mathbb{E}_3$ -algebra in $\text{Fil}(D(\mathbb{F}_2))$. Lemma 12.4 identifies its underlying filtration and its filtration-zero comparison with the ordinary cellular Tate filtration. The assumed compatibility at zero then contradicts Corollary 12.2. $\square$

Remark 12.6. Under the filtration-zero compatibility of Proposition 12.5, the obstruction excludes $\mathbb{E}_n$ refinements for every $n \geq 3$, since each restricts to an $\mathbb{E}_3$ refinement. The detecting operation is Q_2 , which first exists at the $\mathbb{E}_3$ level.

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